



DEPARTMENT OF ENGINEERING MATHEMATICS

London Sport 2: Forecasting Physical Activity Participation, Volunteering and Activity Patterns in London

A Machine Learning Analysis of the Active Lives Adult Survey

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of Master of Science in the Faculty of Engineering.

Friday 11th September, 2026

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Declaration

This dissertation is submitted to the University of Bristol in accordance with the requirements of the degree of MSc in the Faculty of Engineering. It has not been submitted for any other degree or diploma of any examining body. Except where specifically acknowledged, it is all the work of the Author.

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Ethics Statement

This project fits within the scope of ethics application 15208, as reviewed by our supervisor, Dalila O'Grady.

Declaration of Third-Party AI Use

Contributor: [Siyan Xin]

Scope of declaration: Research Question 3 – Indoor and Outdoor Participation Forecasting

Third-party AI was used in the following ways within this project. For my individual contribution to Research Question 3, OpenAI ChatGPT was used only for technical support during code development. It was used to suggest limited example Python scaffolding for parts of the data-processing and modelling workflow and to provide debugging advice in response to code errors and error messages. I subsequently adapted, revised, and integrated any relevant suggestions into my own implementation.

I independently made and justified the methodological decisions used in Research Question 3, including the Months–Days participation rule, the rolling-origin validation design, the use of Total Variation and mean absolute error for model selection, the candidate model and hyperparameter settings, the additive log-ratio transformation, the zero-replacement sensitivity analysis, and the interpretation of indoor exposure. I ran the analyses myself, checked the outputs against the source data and audit files, and verified that the resulting probabilities, forecasts, tables, and figures were internally consistent.

ChatGPT was not used to generate or rewrite substantive paper text, It was not used to fabricate data or experimental results. I wrote all dissertation text related to Research Question 3.

I have reviewed and understand every part of the submitted Research Question 3 code and analysis. I take full responsibility for the methodological choices, implementation, reported results, written discussion, and conclusions in my individual contribution.

Contributor: [Yifeng Mao]

Scope of declaration: Research Question 1 – Physical Activity Trends, Borough Differences and Forecasting

AI tools were used on a limited basis to support the technical development of my individual contribution to Research Question 1. OpenAI ChatGPT was used to clarify statistical and machine-learning concepts and to provide small example code snippets. Any relevant suggestions were independently reviewed, adapted, and verified before use.

I independently made the methodological decisions, including the forecasting validation design, model-selection procedure, clustering strategy, and interpretation of the spatial and temporal results.

ChatGPT was not used to generate or rewrite substantive dissertation text, fabricate data or results, generate citations, or determine the conclusions of the analysis. I understand and take full responsibility for all code, methods, results, discussion, and conclusions included in my individual contribution.

Contributor: [Jingyi Hua]

Scope of declaration: Research Question 2 – Physical Activity Forecasts by Age and Disability Group

OpenAI ChatGPT was used to support my individual contribution to Research Question 2 in two areas. During technical development, it was used to clarify statistical and machine-learning concepts, provide small example Python code or modelling approaches, and assist with debugging when code errors or unexpected outputs occurred. It was also used for limited language support, including grammar correction, wording refinement, and translation. Any relevant suggestions were independently reviewed, adapted, and verified before use.

I independently made and justified the methodological decisions used in Research Question 2, including the construction of the eight forecasting tasks, the rolling-origin validation design, the choice of Total Variation and mean absolute error as the primary evaluation metrics, the additive log-ratio transformation

for compositional outcomes, the recursive forecasting and historical backtesting design, the small-sample reliability and shrinkage checks, and the long-horizon cross-model sensitivity analysis. I ran the analyses myself and checked the resulting forecasts, tables, figures, and diagnostic outputs for consistency with the underlying data and modelling design.

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I have reviewed and understand the code, methods, results, and written analysis included in my individual contribution. I take full responsibility for the methodological choices, implementation, reported results, discussion, and conclusions for Research Question 2.

Contributor: [Shuhan Zhao]

Scope of declaration: Research Question 4 – Frequent Sport Volunteering: Historical Analysis and Forecasting

In the area for which I was personally responsible, OpenAI ChatGPT was mainly used to provide technical support in three areas: firstly, to explain concepts relating to machine learning and to provide sample code for clarification; secondly, to provide debugging suggestions when errors occurred in the code; thirdly, to provide a coding framework for generating certain graphs. I independently assessed, adapted and verified all relevant recommendations before integrating them into my own implementation. The methodological decisions regarding model development, validation, predictions and sensitivity analyses for RQ4 were all made and justified by me independently.

ChatGPT was not used to generate or rewrite the content of this dissertation, nor to falsify data or experimental results, nor did it play any part in establishing the conclusions of the analysis. All the text in this section was written by me. I have reviewed and understood every part of the submitted code and analysis, and I take full responsibility for the choice of methodology, the implementation and the reporting of the results.

Abstract

Insufficient physical activity and unequal access to participation opportunities pose important challenges for public health and sport planning in London. This study developed a unified, survey-weighted framework to analyse and forecast physical activity participation, activity settings and frequent sport volunteering at London-wide and borough levels. It used eight waves of Sport England’s Active Lives Adult Survey from 2015/16 to 2022/23, while the indoor/outdoor and volunteering components used the final six waves. Training, validation and independent test periods were defined sequentially. Naive persistence, Ridge Regression, Random Forest and Gradient Boosting were compared, complemented by spatial analysis, clustering, recursive backtesting and sensitivity analysis.

- Historically, the proportion of London residents classified as Active increased from 64.6% to 66.4%, while the Inactive proportion also rose from 22.2% to 23.8%. The formal annual forecasts did not identify a clear long-term direction for London overall or Inner/Outer London. As a supplementary scenario, the annualised quarterly forecast projected an approximately 2.0-percentage-point increase in the Active proportion by 2030/31, alongside persistent inequalities between boroughs.
- Forecast trajectories stayed stable for most age groups. The oldest age groups and the limiting-disability group kept lower predicted Active rates through the whole forecast period. The backtest results were strong for most tasks. Five of the eight tasks beat the Naive baseline at every tested horizon. Some tasks improved by more than 20%. Two disability composition tasks also showed clear gains on the independent test set, close to 29% and 26%.
- For indoor/outdoor activity, three rolling-origin folds selected Random Forest at both annual and monthly frequencies. On the untouched 2022/23 test wave, annual and monthly Total Variation improved over Naive persistence by 18.2% and 17.6%, respectively. The projected 2028/29 indoor exposure share was 56.5% under persistent COVID legacy and 58.0% under legacy recovery; Active respondents had a lower observed indoor exposure share than the other activity-level groups.
- For frequent sport volunteering, three task specific models succeeded in reducing the effective- n -weighted forecast error by 17.0–24.5% compared with the naive persistence method in the unprocessed 2022/23 test. In the six-year forecasts, the proportion of voluntary work in the Active groups was higher on average, while the projected Active-Inactive gap varied across age groups.

Overall, the findings provide evidence to support London Sport’s borough prioritisation, inclusive activity provision and sport-volunteering planning. However, the short historical series, repeated cross-sectional survey design, sparse samples in some subgroups and reliance of recursive forecasts on generated values limit forecast precision. Long-horizon forecasts should therefore be interpreted as conditional planning scenarios rather than causal estimates or fixed policy targets, and should be updated as new survey waves become available.

Supporting Technologies

- Sport England's *Active Lives Adult Survey* was used as the main respondent-level data source for all four research questions.
- Contextual borough-level data from the Office for National Statistics and the Metropolitan Police were used in the geographical and explanatory analysis.
- Python and Jupyter Notebook were used to develop and execute the data-processing, statistical-analysis and forecasting workflows.
- *Pandas*, *NumPy* and *Pyreadstat* were used to extract, harmonise, validate and aggregate the survey data across multiple years.
- *Scikit-learn* was used for preprocessing, clustering, dimensionality reduction and the implementation and evaluation of the forecasting models.
- *Statsmodels* and *SciPy* were used for statistical modelling, time-series decomposition and supporting statistical tests.
- *GeoPandas*, *Matplotlib* and *Seaborn* were used for geographical processing and the production of maps and statistical figures.
- Git and GitHub were used for collaborative version control, code sharing and integration of work completed by the project team.
- $\text{\LaTeX}$ was used to prepare and format the dissertation through the online collaborative platform *Overleaf*.

Notation and Acronyms

ALR	:	Additive log-ratio transformation
ARI	:	Adjusted Rand Index
FDR	:	False discovery rate
GEE	:	Generalised estimating equations
MAE	:	Mean absolute error
ONS	:	Office for National Statistics
PCA	:	Principal component analysis
RMSE	:	Root mean squared error
RQ	:	Research question
STL	:	Seasonal-trend decomposition using LOESS
TV	:	Total variation distance

w_i	:	Survey weight for respondent i
$n_{\text{eff},g}$	:	Kish effective sample size for group g
p_j	:	Proportion belonging to outcome category j
Q_{ij}	:	Participation indicator for respondent i and activity j
z_j	:	Additive log-ratio coordinate
R^2	:	Coefficient of determination

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Chapter 1

Introduction

1.1 Project Background and Research Motivation

Insufficient physical activity is a persistent public health and social planning issue. A global analysis of 358 population-based surveys involving approximately 1.9 million adults found that insufficient activity remained widespread, with trends differing substantially across regions and population groups [1]. These differences affect not only population health, but also equity in opportunities to lead active lives and the allocation of local sport and community provision. Historical monitoring can describe current patterns, but cannot determine whether geographical and demographic disparities will persist, narrow, or widen. Medium and long term planning therefore requires forward-looking evidence generated by models evaluated out of sample, rather than relying only on current averages or simple extrapolation.

This need is particularly important in London because socioeconomic inequalities are associated with unequal physical-activity opportunities [2]. Neighbourhood conditions also contribute to variation within deprived London communities [3]. London-wide averages may therefore conceal local needs, although excessive disaggregation creates sparse samples and unstable estimates. Reliable forecasts can help London Sport identify priority boroughs and groups, plan indoor provision and outdoor spaces, support volunteering, and target further data collection.

A single headline participation rate is also insufficient. Activity profiles vary substantially with age [4]. Adults with limiting disabilities report lower participation and shorter activity duration [5]. An improving aggregate rate could therefore conceal groups whose opportunities are deteriorating, so forecasts must examine activity status and specific participation across population groups.

Where activity takes place introduces another planning dimension. Urban green-space interventions can affect activity opportunities and equity [6]. Investment in active-travel infrastructure can also change physical activity and travel behaviour [7]. Sport participation is associated with sport volunteering, which supports clubs and community programmes [8]. Forecasting settings and volunteering alongside overall participation therefore gives London Sport a broader account of future demand.

These practical needs motivate the project *London Sport 2: Using Machine Learning to Forecast Physical Activity Participation, Volunteering and Activity Patterns in London*. The project moves beyond a one-off historical description by producing comparable forward-looking evidence across geographical areas, population groups, activity settings, and forms of community engagement.

1.2 Research Aim, Questions and Scope

The overall aim is to provide London Sport with effective and interpretable evidence on how physical activity participation, activity settings, and frequent sport volunteering may develop across London, thereby supporting future planning and resource prioritisation. The project addresses four research questions:

RQ1 (core): How are physical activity levels in London overall, Inner London, Outer London, and the 32 London boroughs expected to change over the next eight years?

RQ2 (core): How are physical activity levels and participation in specific activities expected to change across age and disability groups over the next eight years? Which activities are predicted to have the highest participation rates within each group?

RQ3 (expansion): How might indoor and outdoor activity participation in London overall and across the boroughs change over the next six years? Do indoor and outdoor activity patterns differ between groups with different physical activity levels?

RQ4 (expansion): How will frequent sport-volunteering rates change over the next six years across London boroughs, age groups, and activity-level groups? Does the forecast gap between activity-level groups vary by age?

The study uses Sport England’s Active Lives Adult Survey, which covers adults aged 16 and over in England and supports local-authority estimates [9]. Its geographical scope comprises London overall, Inner London, Outer London, and 32 boroughs. The City of London is excluded from borough-level comparisons because of its distinctive administrative status and sample characteristics. RQ1 and RQ2 use eight survey waves from 2015/16 to 2022/23 and generate eight-year forecasts; RQ3 and RQ4 use six waves from 2017/18 to 2022/23 and forecast from 2023/24 to 2028/29.

1.3 Research Positioning, Core Challenges and Achievements

Existing evidence links physical activity to socioeconomic and neighbourhood conditions [2, 3]. Age and disability are also associated with different participation profiles [4, 5]. Environmental opportunity and volunteering add distinct planning dimensions [6, 8]. However, these dimensions are rarely forecast together for local planning in London. This project connects them through a common survey source and chronological evaluation framework. Persistence, regularised and nonlinear models are compared using out-of-sample origins because forecasting methods should be judged on observations later than their training data [10].

Three challenges shape the analysis. First, only six or eight annual observations support forecasts of equal length. COVID-19 also changed activity opportunities and behaviour, making temporary disruption difficult to separate from longer-term change [11]. Second, questionnaire fields must be harmonised while borough-group intersections create sparse cells. Third, Active Lives is a self-reported repeated cross-sectional survey and does not follow the same residents over time [12]. The forecasts are therefore conditional scenarios rather than causal estimates or fixed policy targets.

The project’s aims, objectives, and achievements are summarised as follows:

- **Aim:** To provide London Sport with forward-looking evidence across geographical areas, population groups, activity settings, and sport volunteering.
- **Objective 1 and achievement:** Completed multi-scale trend, spatial, clustering, and forecasting analysis, identifying persistent differences across London boroughs.
- **Objective 2 and achievement:** Produced age and disability forecasts, highlighting the future trajectories of limiting-disability and oldest-age groups as particular concerns.
- **Objective 3 and achievement:** Constructed comparable indoor/outdoor measures, separated the two meanings previously combined as “neither”, and produced 2023/24–2028/29 forecasts under two COVID-19 scenarios. Both annual and monthly Random Forest models improved on Naive in the untouched 2022/23 test, while activity-level groups showed different indoor exposure balances.
- **Objective 4 and achievement:** Forecast frequent sport volunteering across age and activity-level groups, identifying age-related variation in the projected Active–Inactive gap.
- **Cross-cutting achievement:** Used chronological validation, naive baselines, survey-weighted estimation, and sensitivity checks to distinguish out-of-sample evidence from longer-term scenarios requiring future validation.

Chapter 2

Background

This chapter introduces the Active Lives Adult Survey and the main evidence behind the four research questions. It also explains the models and evaluation ideas used later in the study. Previous research has documented clear differences in physical activity across places and population groups. The studies reviewed here provide less evidence on local multi-year forecasting, especially when borough, age, disability, activity setting and volunteering are considered within one framework.

2.1 The Active Lives Adult Survey

The Active Lives Adult Survey has measured sport and physical activity among adults aged 16 and over in England since November 2015. Sport England uses the survey to report results at national, regional and local-authority levels [9]. Its repeated annual waves make it suitable for studying changes over time and for comparing London boroughs within one survey system.

The survey classifies respondents according to weekly moderate-intensity activity. Adults completing at least 150 minutes per week are classified as Active. Those completing 30–149 minutes are Fairly Active, and those completing fewer than 30 minutes are Inactive [13, 14]. These three groups provide a broad measure of physical activity level.

Active Lives also records participation in individual activities. The survey includes information on participation during the previous 12 months, participation frequency, activity setting, age, disability and sport volunteering [12, 13, 14, 15]. These variables allow the same data source to support both broad activity-status analysis and more detailed questions about what people do, how often they participate and which groups show different patterns.

The survey uses online and paper questionnaires. Some variables are available only through particular questionnaire routes, and the available weights depend on the type of analysis [12, 15]. This matters because annual estimates and more detailed activity measures may require different survey weights. The specific choices used in this study are described in Chapter 3.

Each annual wave is a new cross-section of the population. Active Lives does not follow the same respondents from one year to the next [12, 15]. Changes across survey years therefore describe population-level patterns at different points in time.

The survey is completed by respondents online or through a paper questionnaire, and activity information is based on their reported participation [12, 15]. This means that differences in recall or interpretation may still affect the recorded values. Survey weighting improves population representation, but it does not remove this type of measurement uncertainty. The issue becomes more important when the data are divided into small borough or demographic groups.

Weighted population size and respondent count also describe different features of an estimate. A cell may represent a large population but still contain only a small number of respondents. For this reason, the study also considers Kish effective sample size when respondent-level weights are available. It is used as an indicator of historical sample support alongside other diagnostics. Related small-area methods use multilevel time-series models to share information across domains and time when direct estimates are based on limited samples [16].

2.2 Evidence Underlying the Four Research Questions

The four research questions examine different parts of physical activity planning in London. RQ1 focuses on geographical and temporal differences. RQ2 examines age, disability and participation in specific activities. RQ3 studies indoor and outdoor activity settings. RQ4 examines frequent sport volunteering. These questions are linked because activity can vary across place, population group and type of participation.

Physical activity differs across geographical areas. Research using data from more than one million adults in England found clear socioeconomic differences in physical inactivity [2]. London-specific research also found links between physical activity and both individual and neighbourhood conditions in deprived communities [3]. London-wide averages can therefore hide important local differences.

The local environment can shape the opportunities available for physical activity. Research on urban green space has found links between environmental changes and physical activity, health and wellbeing [6]. The ENABLE London study also examined how changes in the built environment could support more active lifestyles [17]. In Outer London, investment in active-travel infrastructure was associated with increases in walking and active travel [7]. These findings support borough-level analysis because access to activity opportunities can differ across London.

A borough's current activity level does not show its full history. Two boroughs may have similar values in one year but follow different paths across several years. Neighbouring boroughs may also share transport systems, facilities or socioeconomic conditions. RQ1 therefore examines both activity levels and changes over time, together with spatial and clustering patterns.

The COVID-19 period adds an unusual break to the historical series. Active Lives data showed substantial changes in physical activity during the first national lockdown [18]. A wider review also reported changes in physical activity and sedentary behaviour during lockdown, although the size of these changes differed across populations [11]. The historical data used in this study therefore contain both longer-term patterns and a period of major disruption.

Age is closely linked to physical activity. Activity profiles differ across age groups, and the types of activity people take part in also change across the life course [4]. Work, retirement and changes in daily routines can affect the time and opportunities available for activity [19]. These differences mean that one population-wide activity rate may hide important age patterns.

The situation becomes more complex among older adults. Research using Active Lives data found that sport participation among older adults was related to age, deprivation and perceived opportunities [20]. A review of adults aged over 70 also identified health concerns, fear of falling and environmental barriers as common problems, while social support and accessible facilities could help participation [21]. Separate age groups are useful because the conditions affecting a person aged 55 may differ from those affecting someone aged 85.

The oldest age groups also create a practical data problem. Borough-level samples become smaller as the data are divided by age. This can make annual observations more variable even when the underlying population pattern changes only slightly. RQ2 therefore keeps the age groups separate and also examines historical sample support when interpreting their forecasts.

Disability adds another source of variation. The World Health Organization describes disability as an interaction between health conditions, daily functioning and the surrounding environment [22]. The barriers faced by disabled people can therefore depend on access, mobility, support and the type of activity available.

Global research has found substantial differences in physical activity participation between disabled and non-disabled populations [23]. Other reviews have identified barriers related to facilities, transport, social support and the organisation of mainstream sport [24, 25]. These barriers do not affect every disabled person in the same way, so one broad disability category can hide important differences.

Active Lives provides evidence that is closely related to the outcomes used in this study. Carr et al. reported differences in physical activity type and duration between adults with limiting disabilities, non-limiting disabilities and no disability [5]. Common activities such as walking, swimming and fitness remained important, but participation levels differed across disability groups. RQ2 therefore examines both broad disability status and more detailed disability indicators.

Similar reliability problems also appear in the disability analysis. Some respondents report more than one disability, so the detailed disability indicators can overlap. Dividing these groups further by borough reduces the number of observations available for each cell. These estimates are therefore interpreted together with their historical sample support.

RQ2 uses four related outcomes to describe these patterns. Overall activity composition shows how

participation is divided between Active, Fairly Active and Inactive. Activity-specific composition applies the same three-level Active–Fairly Active–Inactive structure separately to each retained activity. Past-12-month participation describes broad participation reach, and regular participation gives information on more sustained engagement. These outcomes separate general activity level from participation in specific activities.

Activity ranking adds a practical view for planning. The ranking is based on predicted participation rates. It shows which activities are expected to have the highest participation within each group under the fitted model and observed history. The ranking is not a measure of stated preference. When the first and second predicted rates are close, a small forecasting error may change their order. The difference between the two predictions is therefore also retained.

Activity setting creates another planning question. Outdoor spaces can support walking, cycling and informal activity, while indoor facilities can provide organised sessions and equipment. Research on green space and active travel shows that access to outdoor opportunities can be linked to physical activity [6, 7]. Access to indoor and outdoor settings also changed during the COVID-19 period [11].

Indoor and outdoor participation can overlap. The same respondent may report activity in both settings during one survey period. RQ3 therefore studies the balance between recorded settings and how that balance changes over time. This allows the analysis to keep participation in both settings within the same framework.

Sport volunteering supports clubs, events, coaching and community programmes. The likelihood of volunteering differs across demographic and socioeconomic groups [26]. Research also shows that previous involvement in organised sport is related to sport volunteering [27].

Participation and volunteering are connected forms of involvement in sport. Dawson and Downward found an association between sport participation and sport volunteering in UK survey data [8]. Earlier volunteering behaviour can also be linked to future volunteering [28]. These patterns support the inclusion of activity level in the volunteering analysis.

Local volunteering estimates can become unstable when borough, age and activity level are combined. Recent work using Active Lives has also highlighted the difficulty of producing detailed local volunteering estimates from survey data [29]. RQ4 therefore keeps small-cell information visible when comparing volunteering patterns across groups.

2.3 Forecasting Models, Clustering and Evaluation

The forecasting tasks use different outcomes, but they face the same basic problem. Only six or eight annual survey waves are available, while the final forecasts extend several years into the future. A useful model needs to make use of the available history without fitting short-term noise too closely.

The Naive model provides a simple baseline. It carries forward the most recent observed value and shows whether a fitted model adds useful predictive information. Ridge Regression provides a regularised linear model and can reduce unstable coefficient estimates when predictors are related [30].

Random Forest and Gradient Boosting provide nonlinear alternatives. Random Forest combines many decision trees and can represent interactions without specifying them in advance [31]. Gradient Boosting builds trees in sequence, with later trees focusing on prediction errors left by earlier ones [32]. These models can capture more flexible relationships, but the short historical series limits how much complexity the data can support.

The study compares the candidate models using later-year predictive performance. This gives a more useful test for forecasting than relying on how closely a model fits the data used to train it. The same principle also keeps the Naive baseline important, because a complex model only adds value when it improves predictions on unseen periods.

RQ1 also uses clustering to describe differences between boroughs. Clustering and forecasting serve different purposes. Forecasting estimates later outcomes, while clustering identifies boroughs with similar observed characteristics or historical paths.

K-means groups observations by reducing variation within each cluster [33]. Ward hierarchical clustering builds groups step by step while limiting the increase in within-cluster variation [34]. These two methods provide different views of borough structure and allow the stability of the resulting groups to be checked.

Several measures are used to assess the clustering results. The silhouette score describes how well observations fit within their assigned cluster compared with nearby clusters [35]. The Adjusted Rand Index measures agreement between different cluster solutions and is useful for checking stability [36].

Moran’s I examines whether similar values are spatially concentrated [37]. Each measure describes a different part of the borough structure.

Survey-weighted estimates also require information about sample support. When respondent-level survey weights are available, the Kish effective sample size is calculated as [38]

$$n_{\text{eff},g} = \frac{\left(\sum_{i \in g} w_i\right)^2}{\sum_{i \in g} w_i^2}, \quad (2.1)$$

where g is a grouped cell and w_i is the survey weight for respondent i . When survey weights within a cell are similar, n_{eff} stays close to the raw sample size. Unequal weights reduce the effective amount of information. In this study, n_{eff} is used to support weighting, diagnostics and small-sample sensitivity checks.

Overall physical activity status has a different modelling problem. Active, Fairly Active and Inactive form a composition because the three proportions are non-negative and sum to one. Predicting each proportion separately can produce values that do not satisfy this structure. Log-ratio methods provide a way to model these proportions in an unconstrained space and have been applied in physical-activity research involving compositional outcomes [39, 40].

RQ2 uses Active as the reference category in the additive log-ratio transformation:

$$z_1 = \log\left(\frac{p_{\text{Inactive}}}{p_{\text{Active}}}\right), \quad z_2 = \log\left(\frac{p_{\text{Fairly Active}}}{p_{\text{Active}}}\right). \quad (2.2)$$

The inverse transformation converts the forecasts back to valid proportions. A small numerical replacement is required when one of the observed proportions is zero. The sensitivity of the forecasts to this numerical choice is evaluated later in the study.

Forecast evaluation also needs to respect time order. A random train–test split could allow information from later survey years to influence predictions for earlier periods. Forecast evaluation is stronger when each model uses only information that would have been available before the forecast year [10].

The study therefore uses time-ordered validation. Earlier survey waves are used to fit the model, later waves are used for validation, and the final 2022/23 wave is kept for independent testing. Where enough historical observations are available, expanding-window rolling-origin validation provides several forecast origins. This reduces dependence on one validation year and keeps the evaluation close to the real forecasting task.

The independent test has a different role from validation. Validation is used to select the model and its hyperparameters. The test wave measures generalisation after these decisions have been fixed. This separation reduces the risk of choosing a model because it happened to perform well on the final observed year.

Multi-year forecasting creates an additional challenge because later outcomes are not yet observed. Recursive forecasting uses an earlier prediction to construct the input required for the next forecast period. This produces one continuous future path, but later predictions depend increasingly on earlier model outputs. Research on multi-step forecasting shows that forecast performance can change as the horizon increases [41]. Historical recursive backtesting is therefore used where the available survey history allows it.

The main evaluation metric depends on the type of outcome. Total variation distance is used for compositional outcomes:

$$TV(\mathbf{p}, \hat{\mathbf{p}}) = \frac{1}{2} \sum_{k=1}^K |p_k - \hat{p}_k|. \quad (2.3)$$

TV measures the difference between the observed and predicted composition across all categories. A smaller value means that the predicted composition is closer to the observed one.

Mean Absolute Error is used for single-rate outcomes. MAE remains on the original proportion scale, so it gives a direct measure of the average absolute forecasting error. RMSE and out-of-sample R^2 are used as supporting diagnostics where needed. RMSE places more weight on large errors, and out-of-sample R^2 describes how much variation in unseen data is reproduced by the model.

Chapter 3

Execution

3.1 Common Analysis Methods

The four research questions use different outcome variables and units of analysis but share a common data preparation framework and an overall time-series forecasting design.

3.1.1 Common Data Preparation and Integration

This project uses the original individual-level data from Sport England’s Active Lives Adult Survey, covering eight survey years from 2015/16 to 2022/23. The survey covers adults aged 16 and over in England. Across England, each survey year includes approximately 175,000 respondents, recording participation in sport and physical activity together with relevant demographic information.

First, the data for each survey year were restricted to London’s 32 boroughs. The City of London was excluded because its sample structure differed from that of the other boroughs. Variables whose names changed across survey years were harmonised, and the same rules for variable selection, missing-value handling, and output formatting were applied to all survey years.

The retained variables included respondent identifiers, survey time, borough, Inner/Outer London classification, demographics, activity level, sport volunteering, survey weights, and specific activity variables. After harmonisation, the eight-year dataset contained 135,497 respondent records available for annual analysis, of which 134,916 also contained the time information required for monthly and quarterly analyses. The six-year dataset covering 2017/18 to 2022/23 contained 96,629 records.

Second, two respondent-level versions of the data were created. The eight-year version was used for RQ1 and RQ2 and retained 124 activities available across all survey years. The six-year version used a broader dictionary of 179 activities from 2017/18 to 2022/23 to support the analysis of activity settings in RQ3 and sport volunteering in RQ4. The two data versions were retained to balance comparability over a longer period against the richer set of activity variables available in later survey years.

Third, in both data versions, each row represented one respondent. The initial fields recorded the survey year, borough, demographic characteristics, overall activity level, and survey weight, while the subsequent fields recorded participation, participation frequency, activity level, and activity setting information for each specific activity. Standardised activity-variable suffixes allowed the same activity to be matched across different variable groups and survey years.

Finally, codes representing no response, not applicable, or questionnaire routing were treated as missing values, whereas valid zero values indicating actual non-participation were retained. If an activity was not surveyed in a particular survey year, the corresponding field remained missing rather than being recoded as non-participation. If information about a particular activity was missing for a respondent, that respondent was excluded only from the calculation for that activity, while their other valid information remained available for analysis. This approach avoided misinterpreting non-responses as non-participation while retaining valid data.

3.1.2 Common Forecasting and Evaluation Process

The common forecasting framework consists of four main stages. First, the candidate model set comprises a Naive baseline based on the previous survey year, Ridge Regression, Random Forest, and Gradient

Boosting. The Naive model carries forward the observation from the previous survey year or the corresponding period in the previous survey year. The other models use task-specific predictors constructed from earlier observations, time indicators, and relevant geographical or population-group information. Any lagged or rolling predictors are calculated within the relevant geographical or population-group series and use only information available before the forecast origin.

Second, model validation preserves the temporal order of the data. RQ1 and RQ2 use eight survey years from 2015/16 to 2022/23 and employ four expanding-window, one-year-ahead validation folds: data from 2015/16–2017/18 are used to forecast 2018/19, data from 2015/16–2018/19 to forecast 2019/20, data from 2015/16–2019/20 to forecast 2020/21, and data from 2015/16–2020/21 to forecast 2021/22. RQ3 and RQ4 use six survey years from 2017/18 to 2022/23 and employ three validation folds: data from 2017/18–2018/19 are used to forecast 2019/20, data from 2017/18–2019/20 to forecast 2020/21, and data from 2017/18–2020/21 to forecast 2021/22. The four and three fold designs used the available pre-test years while retaining an initial training period and reserving 2022/23 for independent testing. Expanding-window validation preserves the information that would have been available at each historical forecast origin while allowing the training sample to increase progressively over time.

Third, the primary evaluation metric is selected according to the form of the outcome variable. For the compositional outcome consisting of Active, Fairly Active, and Inactive, TV is used as the primary metric because it evaluates the complete composition rather than a single category. For individual participation rates and sport-volunteering rates, MAE is used as the primary metric because it remains on the original proportion scale. RMSE and out-of-sample R^2 are reported as supplementary metrics to identify larger forecasting errors and evaluate performance on unseen data, respectively. The task-specific primary metric is used to compare candidate models and hyperparameter settings across the rolling-origin validation folds.

Finally, the 2022/23 survey year is held out as independent test data until model and hyperparameter selection has been completed. It therefore does not influence the choice of model family or hyperparameters and provides an independent assessment of point-forecast generalisation performance. After testing, the selected model is refitted using all available historical survey data to generate future forecasts. Multi-year forecasts are generated recursively because observed lagged outcomes are unavailable beyond 2022/23. The prediction for one future survey year is therefore used to construct the lagged input required for the next. This approach generates a continuous multi-year forecast path, although forecasting errors may accumulate as the forecast horizon increases. All analyses use fixed random seeds to ensure reproducibility.

3.2 Physical Activity Trends, Borough Differences and Forecasting

3.2.1 Data and Multilevel Analysis

RQ1 uses the eight-year dataset generated through the common data preparation process and constructs analytical datasets at the London-wide, Inner/Outer London, borough, and respondent levels. In addition to the Active Lives data, the study incorporates three borough-level contextual data sources: ONS mid-year population estimates and population change data, residence-based weekly earnings from the ONS Annual Survey of Hours and Earnings, and monthly borough-level crime records published by the Metropolitan Police [42, 43, 44]. These sources were used to derive borough-level indicators of population structure and change, migration, earnings, and selected population-standardised crime rates.

The first analysis examines temporal changes at the London-wide level. The study first compares the annual and monthly proportions of Active, Fairly Active, and Inactive, and then uses STL to decompose the monthly Active series into long-term trend, recurring seasonality, and irregular fluctuations. To describe changes during the pandemic, time is divided into pre-pandemic, strict-restriction, reopening, and subsequent phases, and differences between each phase and the pre-pandemic period are compared using a weighted least-squares model. The model also controls for a linear time trend, survey month, and the proportion of online responses. The resulting estimates are interpreted as adjusted differences between phases rather than causal effects of the pandemic.

The second analysis compares Inner and Outer London. In addition to the overall Active proportion, the study compares the structure of participation in specific activities and uses age standardisation to adjust both areas to the same London age distribution. For specific activities, the raw participation rate is used as the primary outcome, while the base-2 logarithm of the ratio between Inner and Outer London participation rates is calculated as a measure of relative activity concentration. The monthly

Inner-minus-Outer gap was modelled directly using weighted least squares, controlling for a linear time trend, survey month, pandemic phase, and the difference in online-response proportions.

The third analysis examines differences across the 32 boroughs. Annual distributions describe variation across eight years, while a Spearman rank correlation compares borough rankings in 2015/16 and 2022/23. A random-intercept mixed model partitions residual variation, after controlling for survey year, into persistent between-borough differences and within-borough annual variation plus residual noise. Global Moran’s I tests clustering in the 2022/23 Active proportions and eight-year changes, while Local Moran’s I identifies neighbouring high-value and low-value boroughs in 2022/23, with adjustment for multiple comparisons. Finally, the within-between model decomposes demographic, economic, migration, and crime indicators into each borough’s long-term mean and annual deviation from that mean, thereby distinguishing relationships associated with persistent borough differences from those associated with changes within the same borough over time.

The fourth analysis returns to the respondent level to capture participation structures that cannot be shown by borough averages. It calculates the number of activities reported by respondents, participation rates over the previous 12 months, higher-frequency participation rates, and trends across activity categories. Participation breadth used 38 non-overlapping activities selected from the original 124 consistently observed activities. Weighted co-participation correlations were calculated for the eight activities with the highest eight-year mean participation to keep the correlation matrix readable and avoid unstable estimates for rarely reported activities. Age differences were analysed using a weighted binomial GEE model with borough as the clustering unit. After controlling for survey year, Inner/Outer London, questionnaire response mode, and disability status, the results are converted into adjusted probabilities of being Active for each age group.

3.2.2 Borough Clustering and Forecasting

Inner/Outer London provides only a broad geographical classification, so the study further uses structural and trajectory clustering to describe more specific local differences. Structural clustering uses eight-year averages of the Active proportion, limiting disability, population aged 65 and over, relative earnings, net migration, and selected recorded crime. The standardised variables are reduced using PCA, after which K-means and Ward solutions containing two to eight clusters are evaluated using separation, cluster size, and bootstrap stability measures. A simplicity rule selects the smallest near-best solution. Resampling and alternative feature specifications are used to assess membership uncertainty and robustness.

Trajectory clustering applies two approaches to the 2015/16–2022/23 Active series for the 32 boroughs. The first retains the original proportions, so both long-term levels and temporal trajectories affect the clustering distances; the second standardises each borough’s series to a mean of 0 and a standard deviation of 1, removing differences in typical activity levels and comparing only trajectory shapes. Structural and trajectory clustering are then compared in terms of borough membership and stability.

RQ1 produced monthly, quarterly, and annual forecasts at the London-wide, Inner/Outer London, and 32-borough levels, resulting in nine forecasting tasks. Each task simultaneously forecast the proportions of Inactive, Fairly Active, and Active. The monthly and quarterly models used the previous period’s proportions, short-term recent averages, proportions from the corresponding period of the previous survey year, linear trends, and seasonal information. The annual models used the previous survey year’s proportions and linear trends. RQ1 forecast the three proportions directly and adjusted them after forecasting to obtain valid proportions that summed to 1.

For the non-Naive models, training weights were based on the Kish effective sample size, with the influence of extreme weights on model fitting limited. Model selection was based on the mean Kish-effective-sample-size-weighted TV across the shared validation folds. Annual forecasts were treated as the primary results because the research question and the main geographical comparisons were defined at the survey-year level, ensuring consistency between the historical analysis and the future forecasts. Quarterly forecasts provided supplementary evidence on within-year variation, while monthly forecasts were placed in the Appendix because their greater sampling variability, weaker effective-sample-size support, and 96-step recursive horizon made them unsuitable for the main analysis.

The proportion of the population aged 65 and over, the adult population growth rate, and median weekly earnings were each examined in supplementary tests for 2021/22 using a one-year lag, but were not included in the primary models, the independent 2022/23 test, or the final forecasts. Crime indicators were used only for explanatory analysis and clustering. The selected models generated forecasts for 2023/24 to 2030/31. Indicative ranges constructed from historical validation and test residuals were used only to illustrate forecast sensitivity and were not treated as formal statistical prediction intervals.

3.3 Forecasting Physical Activity Participation by Age and Disability Group

RQ2 examines future physical activity across age and disability groups and identifies activities with the highest predicted participation rates. The age strand contains eight groups from 16–24 to 85+, while the disability strand contains three broad disability-status groups and 13 detailed disability indicators. The two strands are modelled separately but follow the same time-ordered forecasting framework.

3.3.1 Data and Outcome Construction

Respondent records were aggregated into annual borough-level panels using the relevant survey weights. Four forecasting tasks were constructed for each demographic strand: overall Inactive–Fairly Active–Active composition, activity-specific composition, past-12-month participation (`MONTHS_12`), and regular participation measured by `DAYS10P60GR`. This gives eight forecasting tasks across the age and disability strands. Activity-specific analysis used the 124 activities retained across all eight survey waves.

Missing responses followed the common missing-data treatment. They were not coded as non-participation and were excluded only from the calculation of the affected outcome. No additional imputation of outcome values was introduced for RQ2.

Raw respondent counts and Kish effective sample size were retained as indicators of historical sample support. These measures were used in diagnostic and supplementary small-sample analyses rather than as automatic exclusion criteria.

The overall and activity-specific activity-level outcomes were treated as compositions and modelled using the additive log-ratio (ALR) transformation, with Active as the reference component. A numerical constant of 10^{-6} was applied to zero components so that the log-ratio transformation could be calculated. Predictions were then returned to the original proportion scale using the inverse ALR transformation, producing non-negative shares that sum to one.

3.3.2 Forecasting Design and Model Selection

Temporal predictors were constructed within each borough and demographic series. The overall composition tasks used one- and two-year lagged outcomes. The activity-specific composition and single-rate tasks used a one-year lag and a two-year historical rolling mean. All fitted models also included a linear time trend and a COVID-period indicator. Ridge Regression additionally included demographic-group-by-time interaction terms.

The COVID indicator marked the 2019/20 and 2020/21 survey waves and was set to zero from 2023/24 onward. The production forecasts therefore do not assume that the pandemic-period indicator continues into the future.

Numerical imputation and categorical encoding were fitted within the training pipeline and then applied unchanged to the corresponding validation or test data. Categorical predictors were one-hot encoded, and categories not observed during training were handled through the encoder’s unknown-category rule. Validation and test observations were scored only when the required lagged predictors could be constructed from information available at the relevant forecast origin.

Model family and hyperparameters were selected using expanding-window rolling-origin cross-validation. RQ2 used four one-year-ahead validation waves: 2018/19, 2019/20, 2020/21 and 2021/22. Performance was averaged across these four origins before the model family and hyperparameters were fixed. TV was used as the primary metric for the two compositional tasks, while MAE was used for `MONTHS_12` and `DAYS10P60GR`. The 2022/23 survey wave was then used once as an independent test and did not alter the selection decision.

A borough-random-intercept mixed-effects model was retained only as a supplementary comparison for the two overall composition tasks.

3.3.3 Recursive Forecasting and Reliability Assessment

After independent testing, the selected specifications were refitted using all observed survey waves from 2015/16 to 2022/23 and used to generate forecasts for 2023/24–2030/31. Forecasting was recursive, so predictions from one future survey wave were reused to construct lagged inputs for the following wave.

Historical recursive backtesting examined how this mechanism behaved over periods for which later observations were available. A model fitted using information up to 2018/19 recursively forecast 2019/20–2022/23, providing horizons H1–H4. Later forecast origins used information up to 2019/20, 2020/21

and 2021/22 and produced progressively shorter paths ending in 2022/23. At each step, the previous prediction was reused as part of the next input. Recursive Naive persistence was evaluated under the same origins and horizons.

A no-look-ahead rule was applied to the COVID indicator. If future COVID status would not have been known at a historical forecast origin, the later observed status was not supplied to the model and the future indicator was set to zero.

The backtest therefore provides direct evidence on recursive forecast behaviour up to H4. It does not validate the specific future years themselves. H4 is represented by only one historical origin, and H5–H8 extend beyond the directly backtested range. The longer-horizon forecasts therefore depend more strongly on the stability of the fitted relationships and on recursively generated inputs.

A supplementary shrinkage analysis examined whether predictions with weaker historical sample support could be stabilised without changing the selected model family or hyperparameters. For compositional outcomes, each borough-level prediction vector was adjusted towards a leave-one-borough-out London reference:

$$\hat{\mathbf{p}}^{adj} = \frac{n_{\text{eff}}^{support}}{n_{\text{eff}}^{support} + \lambda} \hat{\mathbf{p}}^{borough} + \frac{\lambda}{n_{\text{eff}}^{support} + \lambda} \hat{\mathbf{p}}^{London}. \quad (3.1)$$

Here, $n_{\text{eff}}^{support}$ is the median Kish effective sample size from survey waves strictly before the prediction year. The London reference is the equal-weight mean prediction for the same year, demographic group and activity across the other boroughs, excluding the focal borough. For the single-rate outcomes, the same adjustment was applied in scalar form.

The same shrinkage weight was used for all three components of a composition. Because both the borough prediction and the London reference are valid compositions, their convex combination preserves the unit-sum constraint. The adjusted shares were also renormalised to remove small floating-point discrepancies.

The shrinkage strength λ was selected from the prespecified candidate grid $\{0, 2, 5, 10, 20\}$ using rolling validation while the underlying forecasting specification remained fixed. Cells with observed $n_{\text{eff}} < 30$ formed the primary low-support validation subset used for this selection. The observed effective sample size in the prediction year was used only for retrospective validation or test stratification and did not determine the amount of shrinkage applied to that prediction.

3.3.4 Activity Ranking and Robustness Checks

Activities were ranked using predicted MONTHS_12 participation. For each borough, demographic group and future survey wave, the two activities with the highest predicted participation rates were retained together with the difference between their predicted rates. This ranking margin helps distinguish clearly separated rankings from cases in which the two leading predictions are very similar. The ranking describes predicted participation and should not be interpreted as stated preference.

Sensitivity checks examined the effect of the ALR numerical constant, training-weight specification, COVID specification and alternative definitions of low sample support. Residual-based recursive simulations were also used as an indicative uncertainty stress test rather than as calibrated prediction intervals.

A targeted cross-model comparison generated alternative long-horizon paths using Ridge Regression, Random Forest, Gradient Boosting and Naive persistence. It examined the direction of the 85+ Active-rate trajectory and the direction of change in the Active-rate gap between the no-disability and limiting-disability groups. These comparisons were used to identify model dependence in selected long-horizon findings and did not replace the validation-selected primary specifications.

3.4 Indoor and Outdoor Participation Forecasting

RQ3 forecasts indoor/outdoor participation in 2023/24–2028/29 and compares Inactive, Fairly Active and Active respondents using six Active Lives waves, 2017/18–2022/23 [9].

3.4.1 Data Harmonisation and Outcome Construction

The program compares the six CSV headers with the `Stable_composites` sheet in `179.xlsx`. Activity j is retained when MONTHS_12, DAYS10P60GR, INOUTA and INOUTB exist in every wave; 146 of 179 activities qualify. Blanks, non-numeric entries and illegal codes become pandas NaN and are not imputed. Legal

zeroes remain observed: zero in Months or Days fails the participation rule, while zero in a setting field means unrecorded setting.

Qualifying participation for respondent i is defined jointly from Months and Days:

$$Q_{ij} = \mathbf{1}(\text{Months}_{ij} = 1)\mathbf{1}(\text{Days}_{ij} \in \{1, 2\}). \quad (3.2)$$

Setting is evaluated only when $Q_{ij} = 1$. Aggregation over 146 activities gives `indoor_only`, `outdoor_only`, `both`, or `neither_recorded`. This category does not imply physical inactivity: the audit distinguishes no qualifying comparable activity from qualifying participation with neither setting recorded. Appendix C reports both subcategories and the algorithm.

3.4.2 Panel Construction, Modelling, and Evaluation

Using `wt_final` annually and `wt_time` monthly, 96,629 respondent-wave records form 576 annual and 6,847 observed monthly borough-activity-level cells. Missing cells are not zeroes. Partial pooling moves estimates towards the same-period London composition, more strongly when Kish effective sample size is small. Prior strengths represent 5 annual or 20 monthly effective pseudo-observations. Stronger monthly pooling reflects 6,139 monthly cells with $n_{\text{eff}} < 20$, versus five annually. Sensitivity checks use 1 and 10 annually, and 10 and 40 monthly.

Let p denote category proportions. Indoor and outdoor participation rates are $p_{\text{indoor_only}} + p_{\text{both}}$ and $p_{\text{outdoor_only}} + p_{\text{both}}$; they overlap. Their balance is

$$\text{indoor_exposure_share} = \frac{p_{\text{indoor_only}} + p_{\text{both}}}{p_{\text{indoor_only}} + p_{\text{outdoor_only}} + 2p_{\text{both}}}. \quad (3.3)$$

This is the indoor share of recorded setting exposures, not the percentage of respondents participating indoors or their activity duration. Someone in `both` contributes once to each setting, explaining the factor of two in the denominator.

For each borough-activity-level-time cell g , additive log-ratio (ALR) expresses category j relative to `both` [39]. Values below ϵ are raised to ϵ , then rescaled to sum to one:

$$\begin{aligned} \tilde{p}_{gk}^{(\epsilon)} &= \frac{\max(p_{gk}, \epsilon)}{\sum_{\ell=1}^4 \max(p_{g\ell}, \epsilon)}, \quad k = 1, \dots, 4, \\ z_{gj} &= \log \left(\frac{\tilde{p}_{gj}^{(\epsilon)}}{\tilde{p}_{g,\text{both}}^{(\epsilon)}} \right), \quad j \in \{\text{neither_recorded}, \text{indoor_only}, \text{outdoor_only}\}. \end{aligned} \quad (3.4)$$

Primary $\epsilon = 10^{-6}$ is the smallest tested value, avoiding undefined logarithms with minimal change. Values 10^{-5} and 10^{-4} test sensitivity. Inverse ALR ensures non-negative predictions summing to one.

Naive persistence is compared with Ridge regularisation [30]. Random Forest averages regression trees [31]. Gradient Boosting fits sequential learners [32]. Appendix Table C.2 lists parameters selected only through chronological validation.

Three rolling-origin folds preserve temporal order [10]: training on 2017/18–2018/19 validates on 2019/20; training on 2017/18–2019/20 validates on 2020/21; training on 2017/18–2020/21 validates on 2021/22. Total Variation (TV) compares observed p_{gk} with predicted $\hat{p}_{gk}$ across four categories; larger effective samples receive greater weight:

$$\begin{aligned} TV_g &= \frac{1}{2} \sum_{k=1}^4 |p_{gk} - \hat{p}_{gk}|, \\ TV_w &= \frac{\sum_g n_{\text{eff},g} TV_g}{\sum_g n_{\text{eff},g}}. \end{aligned} \quad (3.5)$$

TV checks all four categories; weighted MAE averages absolute errors in indoor and outdoor participation rates. Multiplying MAE by 100 gives percentage points. Both should be small. Candidates are ranked separately by their three-fold mean TV and MAE; the lowest average rank wins, with TV then MAE breaking ties. The untouched 2022/23 test evaluates this fixed choice.

3.4.3 Recursive Forecasting, Sensitivity Analysis, and Reproducibility

After testing, the model is refitted on 2017/18–2022/23. Annual forecasts use the previous year’s composition; monthly forecasts use the same month a year earlier, replacing unavailable observations with predictions. Pandemic disruption motivates scenario comparisons [11]: *persistent legacy* retains post-COVID controls; *legacy recovery* removes them, retaining trend and seasonality. Neither scenario estimates causal effects or uncertainty intervals.

Activity-level contrasts give equal weight to the six observed waves. A Bayesian bootstrap supplies pairwise intervals, Benjamini–Hochberg controls false discoveries, and a weighted model tests the overall activity-level effect after year and borough adjustment. Forecast contrasts remain descriptive. The reproducible run writes 23 CSV files, seven figures, method notes and `run_audit.json`; assertions verify 1,152 annual and 13,824 monthly future rows, six future waves, valid rates and compositions summing to one.

3.5 Frequent Sport Volunteering: Historical Analysis and Forecasting

RQ4 examines how frequent sport volunteering rates may change from 2023/24 to 2028/29 across London boroughs, age groups and activity level groups. And further investigate whether the predicted rate gaps among different activity levels change with age. The analysis follows the common framework described in Section 3.1, but uses three complementary forecasting tasks: `borough-age-activity_level`, `borough-age` and `borough-activity_level`. The first represents the joint age-activity_level problem of RQ4, while the latter two are used to describe borough-level patterns of age and activity levels respectively. The overall analysis includes historical EDA, chronological model development, 6 year recursive forecasting and sensitivity checks for model selection and long-term prediction stability.

3.5.1 RQ4-specific Data Structure and Target

The target for each analytical unit is the survey-weighted frequent sport volunteering rate. The calculation of survey weighting and Kish effective sample size follows the common procedures mentioned in Section 3.1. All three analytical levels were established as complete year-subgroup panels, and each borough-subgroup combination has a corresponding position in the six historical survey years, even when no eligible observation is available in a particular year. Unobserved rates remain missing rather than being coded as zero.

Data retained the Kish effective sample size and sample coverage information for evaluating the reliability of different survey cells. Because the `borough-age-activity_level` hierarchy further divides the samples into finer subgroup cells, substantially reduces the information available within individual cells. Detailed panel coverage and sample support diagnostics are reported in Appendix D.

3.5.2 Historical Descriptive Exploration

Before forecasting, historical descriptive analysis was used to explore historical patterns. First, used the most recent observed wave (2022/23), established separate `borough-age` and `borough-activity_level` baselines to describe the subgroup heterogeneity within each borough before the forecast period. Then, for each of the six historical waves, calculated the borough-median frequent volunteering rate for each age group and activity level, in order to observe the temporal patterns of a typical borough.

To further describe the internal differences in London, the historical linear slope was estimated for each `borough-age` and `borough-activity_level` series, to summarize the changing directions of each subgroup in different boroughs. In addition, Active, Fairly active and Inactive groups were also compared within the same borough, age group and year, to observe whether rate gaps among activity levels have shown age-specific variations in history.

3.5.3 Forecasting Design, Model Selection and Development Robustness

The forecasting features comprised one-year lagged volunteering rate, a lag-missing indicator, a linear time trend, and accordingly the categorical information corresponding to the borough and subgroup of the forecasting task. The lag is constructed on the complete annual grid:

$$x_{g,t}^{\text{lag}} = p_{g,t-1},$$

where g represents the corresponding borough-subgroup series. If the true $t - 1$ observation was missing, the lag remained missing and an earlier observation will not be used instead. For the non-naive models, missing lag performs median imputation within each training sample while retaining the lag-missing indicator. Enable the model to distinguish between observed and imputed lag values.

The model development followed the three-fold rolling validation and the independent test framework mentioned in Section 3.1. The Naive baseline, Ridge Regression, Random Forest and Gradient Boosting were compared on the common lag-available sample. Due to the significant differences in sampling reliability among different RQ4 cells, the effective- n -weighted mean absolute error (MAE) is selected for the formal model. For the frequent/non-frequent volunteering binary outcome, this metric is equivalent to effective- n -weighted Total Variation:

$$\text{MAE}_w = \frac{\sum_g n_{\text{eff},g} |p_g - \hat{p}_g|}{\sum_g n_{\text{eff},g}}.$$

According to this preset criterion, each task independently completed the hyperparameter and model selection. The selections were fixed using the rolling-validation results only and the 2022/23 test could not reopen model selection.

Then, two model development sensitivity designs were used to examine whether the formal selection process overly relies on specific evaluation assumptions. First, the same rolling-validation predictions were re-conducted using unweighted MAE to assess whether the formal model choice depended strongly on effective- n weighting.

Second, a robustness analysis was conducted to explore the potential impact of COVID affected data on model development. Since the three validation waves from 2019/20 to 2021/22 overlap the COVID affected period, and 2019/20 simultaneously involves the transition of volunteering question coding and survey weighting structure, the currently six annual waves cannot reliably separate the COVID effect from the measurement change. First, checked whether the selected model still maintains its predictive advantage in the independent 2022/23 test; Secondly, when recalculating the model ranking, the validation score of 2019/20 is not included. Observe whether the selected model has undergone significant changes. The data of 2019/20 is still retained in the subsequent model training. Therefore, this analysis is only used to test the robustness of the results, rather than to estimate the causal impact of COVID-19.

3.5.4 Six-Year Recursive Forecasting and Forecast-stage Robustness

After model selection, the model family, hyperparameters and specific feature were all frozen and refitted using all the observed targets from 2017/18 to 2022/23. Subsequently, a six-year recursive forecasting was conducted for 2023/24 to 2028/29. The 2023/24 forecast used observed 2022/23 rate as the initial lag where available otherwise the same missing lag handling method as in the model development process was adopted. Starting from 2024/25, the model projection of the previous year was used as the lag input for the next year. Therefore, long-term predictions gradually rely on the previous output of the model itself.

To address whether the rate difference among activity-level changes with the age group, two within-boroughs contrasts were defined in borough-age-activity_level projections:

$$G_{b,a,t}^{AI} = 100 (\hat{p}_{b,a,\text{Active},t} - \hat{p}_{b,a,\text{Inactive},t}), \quad (3.6)$$

$$G_{b,a,t}^{FI} = 100 (\hat{p}_{b,a,\text{FairlyActive},t} - \hat{p}_{b,a,\text{Inactive},t}), \quad (3.7)$$

where b , a and t represent borough, age group and forecast year. Both gaps are expressed as percentage points and are calculated within borough and then summarized by age group.

To assess the robustness of the six-year recursive forecast, a recursive backtest from H1 to H4 was first conducted using historical data, while keeping the formally selected model and parameters unchanged to observe whether the prediction error will continue to expand as the forecast horizon increases. Secondly, the same six-year predictions were generated respectively using the candidate model families and compared cell by cell with the formal model using the median and 90th percentile absolute difference. For borough-age-activity_level task, the magnitudes and direction of Active-Inactive and Fairly Active-Inactive gaps under different models were further compared. These analyses are only used to evaluate the sensitivity of long-term predictions and do not reopen formal model selection.

Chapter 4

Critical Evaluation

4.1 Evaluation of Physical Activity Trends, Borough Differences and Forecasts

4.1.1 London-wide, Inner/Outer and individual-level patterns

London's Active share rose from 64.6% to 66.4%, but Inactive also rose from 22.2% to 23.8% as Fairly Active fell by 3.4 points. The pattern therefore indicates redistribution towards both ends rather than population-wide improvement. Under the STL decomposition, recurring seasonality accounted for 55.4% of monthly variation, compared with 8.0% allocated to the long-term trend. After adjustment for time trend, survey month, and online response proportion, Active participation remained 2.6–4.5 points below the pre-pandemic level across the pandemic phases. This is consistent with wider Active Lives evidence of uneven lockdown-related changes [18], but the estimates here remain associational because residual variation includes survey noise and unobserved shocks.

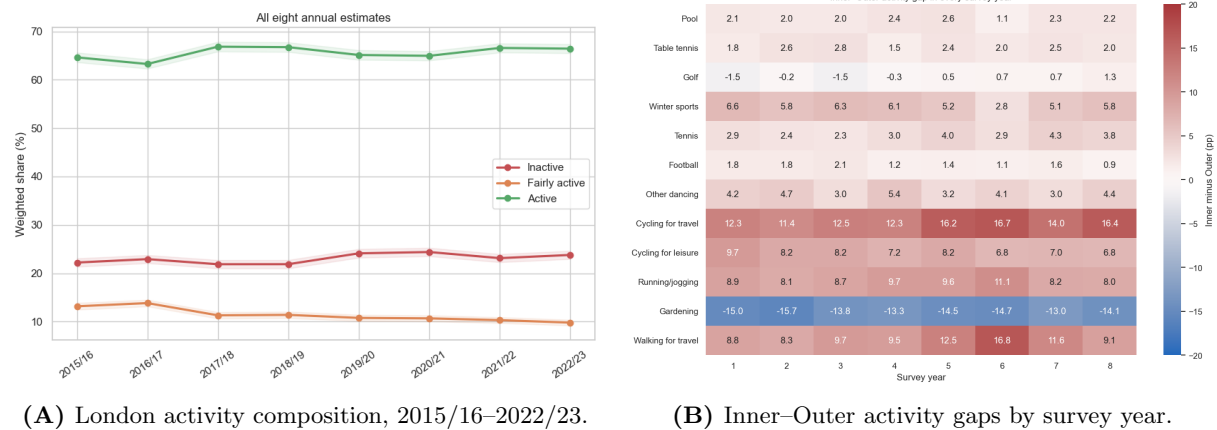


Figure 4.1: Historical activity patterns in London. Positive values in Panel (B) indicate higher participation in Inner London.

Neither Inner nor Outer London fully returned to its pre-pandemic Active level. During the strict-restriction and subsequent periods, Inner London remained 5.5 and 5.6 points below its pre-pandemic level, compared with 3.9 and 3.0 points in Outer London. The Inner-minus-Outer gap narrowed by approximately 1.9 points during strict restrictions, but this change was only borderline significant and did not clearly persist. Across the eight years, Inner London's mean Active share exceeded Outer London's by 5.6 points, falling to 4.5 points after age standardisation. Activity-specific differences were larger: Inner London had 14.0, 10.8, 9.0, and 7.8-point advantages in commuting cycling, commuting walking, running, and leisure cycling, respectively, whereas gardening was 14.3 points higher in Outer London.

These patterns suggest differences in the types of activity opportunities available in Inner and Outer London, rather than only a difference in overall activity levels. High-density areas may provide more

opportunities for activity embedded in daily life through shorter travel distances and public transport connections, while housing and lifestyles in Outer London may support informal activities such as gardening. The environment is only likely to influence participation when it translates into opportunities that are accessible, affordable, and sustainable over time [7, 17].

Individual-level results indicate that overall Active status cannot fully describe behavioural change. The share reporting none of the 38 comparable activities rose from 9.3% in 2015/16 to 14.1% in 2020/21 before easing to 12.5% in 2022/23, while participation in six or more activities fell to 8.1% in 2020/21 and then recovered to 14.4%. In 2022/23, walking for travel reached 70.2%, while 40.4% reported at least two moderate-intensity sessions in the previous 28 days; gardening reached 44.8%, but 15.4% were frequent participants. Several major activity families remained below their 2015/16 levels, while co-participation correlations were modest, with a maximum of 0.24.

After controlling for survey year, Inner/Outer London, response mode, and disability status, the adjusted probability of being Active remained at approximately 65%–70% among those aged 16–74, before falling to 51.6% among those aged 75–84 and further to 33.0% among those aged 85 and over. Therefore, policy should track activity coverage, sustained frequency, and withdrawal risk rather than judging progress by the overall Active share alone.

4.1.2 Borough inequality, spatial patterns and clustering

Borough differences showed clear persistence. The interquartile range of Active levels expanded from 7.7 points to 10.5 points, and the rank correlation between 2015/16 and 2022/23 was 0.71. The multilevel variance model further attributed 75.8% of the remaining variation to persistent differences between boroughs and 24.2% to annual variation within the same borough, measurement error, and other unobserved factors. Active levels in 2022/23 showed significant spatial clustering (Moran’s $I = 0.35$, $p = 0.001$), while the spatial clustering of the eight-year change was only at the borderline level ($I = 0.22$, $p = 0.051$); after adjustment for multiple testing, Wandsworth was identified as High–High, while Newham and Barking and Dagenham were identified as Low–Low.

A borough with long-run relative earnings one standard deviation higher had an average Active proportion 4.75 points higher, but annual changes in earnings within the same borough were almost unrelated to changes in activity, with an estimated association of approximately -0.09 percentage points. Therefore, long-term borough conditions can help explain persistent differences, but cannot simply be translated into the conclusion that short-term changes in earnings lead to activity changes. Adjacent boroughs may also share transport, service, and socioeconomic constraints, so persistently low activity is more likely to require improvements in borough-level conditions for participation rather than being attributed solely to individual choices [2, 3, 45].

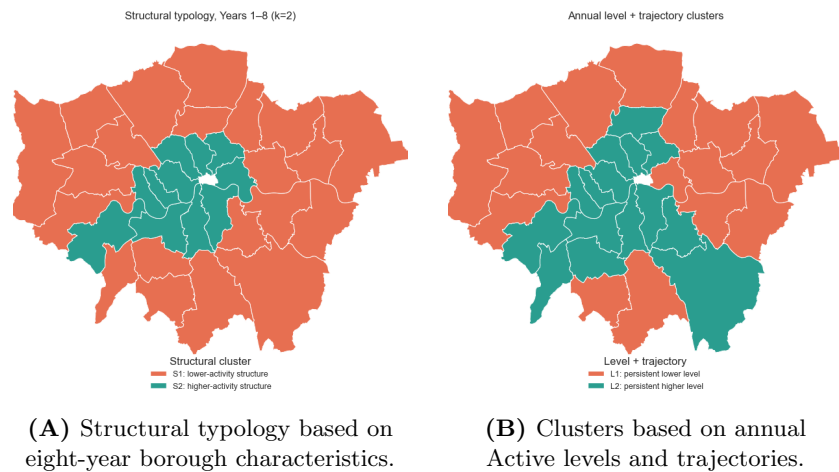


Figure 4.2: Alternative borough classifications based on structural conditions and activity trajectories.

Structural clustering classified 21 boroughs into a lower-activity structural group and 11 into a higher-activity structural group, with a silhouette of 0.310 and a median ARI of 1.000 in the eight-year block bootstrap. PCA retained four principal components, together explaining 91.1% of the variation. After adding participation frequency depth and adult population size, the silhouette dropped to 0.243 and the bootstrap ARI dropped to 0.875, so additional variables did not improve the classification. After remov-

4.1. EVALUATION OF PHYSICAL ACTIVITY TRENDS, BOROUGH DIFFERENCES AND FORECASTS

ing Active, agreement with the main classification remained at 0.874, with only one borough changing category, indicating that the clustering was not simply a transformation of the Active ranking; after removing crime, agreement dropped to 0.549, and four boroughs changed category, indicating that some borderline boroughs were more sensitive to the variable specification. This structural clustering had an ARI of 0.548 with Inner/Outer London, so the two were related but not redundant.

Level-plus-trajectory clustering produced clearer separation: 17 boroughs formed a persistently lower group with mean Active participation of 61.6%, while 15 formed a higher group averaging 70.4% (silhouette = 0.448; median ARI = 0.875). The lower group showed a 1.2-point dip in 2020/21 relative to its earlier mean, followed by a 2.4-point recovery by 2022/23; the higher group was more stable. Its limited agreement with structural clustering (ARI = 0.372) and Inner/Outer London (ARI = 0.371) shows that social structure, activity trajectories, and geography overlap but are not interchangeable. Shape-only clustering was substantially weaker (silhouette = 0.217; median ARI = 0.045), so eight years of data cannot support reliable policy classification based solely on trajectory shape.

These findings also indicate that the value of clustering is not to re-establish borough rankings, but to distinguish different types of problems. Boroughs facing long-term structural constraints and those with persistently weak activity trajectories may require different interventions. The actual participation of residents in activities is also influenced by factors such as opening hours, costs, transportation connections, and cross-administrative area usage.

4.1.3 Forecasting performance and policy implications

In the formal evaluation of the annual forecasts, the London-wide and Inner/Outer London tasks both selected the Naive baseline, with mean validation TV distances of 1.06 and 1.46 percentage points, respectively. The borough-level task selected Ridge Regression, with a mean validation TV distance of 3.82 percentage points. On the held-out 2022/23 test data, the selected annual specifications produced weighted TV distances of 0.66, 0.70, and 3.99 percentage points, respectively. The higher borough-level error is consistent with noisier estimates and more heterogeneous trajectories at finer geographical scales.

Additional tests using alternative models, different hyperparameter settings and random seeds, pooled borough-level changes, and quarterly forecasts aggregated to the annual level did not improve the annual validation errors for London or Inner/Outer London. Each annual series contains only eight time points and includes unusual fluctuations during the pandemic period, making complex models prone to interpreting short-term variation as a long-term trend. The Naive model simply carries forward the composition from the previous year and avoids estimating relationships that are not adequately supported by the available data, resulting in lower validation errors. Its selection does not imply that no future change will occur; rather, it indicates that the available annual data are insufficient to determine the direction of long-term change reliably.

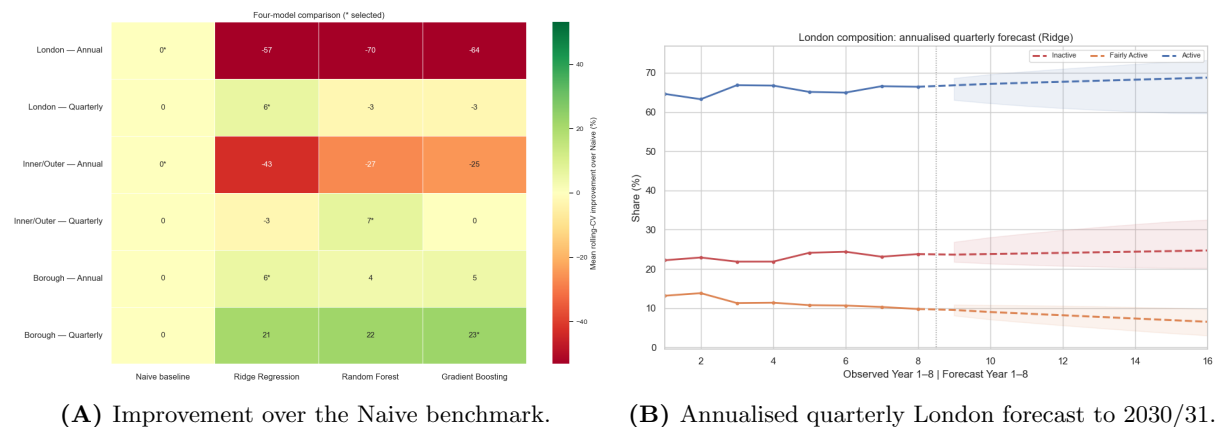


Figure 4.3: Forecast evaluation and the London-wide scenario. Shaded ranges in Panel (B) are sensitivity bands, not prediction intervals.

Quarterly models provided supplementary evidence. Ridge Regression for London and Random Forest for Inner/Outer London improved on their quarterly Naive baselines by approximately 5.9% and 6.9% in rolling-origin validation. Because the formal multi-year annual Naive forecasts remain fixed at the 2022/23 composition, the final results present annualised quarterly scenarios obtained by averaging the

four predictions within each forecast year. These scenarios provide a cautious indication of possible change but do not replace the formal annual model-selection results.

In the supplementary annualised quarterly scenario, London’s Active proportion is forecast to increase by 2.0 percentage points by 2030/31 relative to the average of the final four observed quarters, while Fairly Active is forecast to decrease by 3.2 percentage points and Inactive to increase by 1.2 percentage points. This suggests a possible compositional shift in which the Active and Inactive proportions increase simultaneously. The forecast changes in Active are smaller for Inner and Outer London, at -0.5 and $+0.4$ percentage points, respectively, and therefore do not indicate a shared and clearly defined long-term direction. These results should be interpreted as cautious scenarios of possible change rather than definitive future trajectories.

The final borough-level forecasts use the annual Ridge Regression model directly. By 2030/31, the median borough-level change in Active is forecast to be an increase of approximately 1.1 percentage points, with changes ranging from a decrease of 9.7 percentage points to an increase of 15.3 percentage points. However, the final forecast levels are concentrated between 67.1% and 69.5%, indicating that Ridge Regression pulls boroughs with initially higher and lower activity levels towards more similar values. This convergence partly reflects regularisation and the pooled modelling structure and should not be interpreted as evidence that boroughs will necessarily converge in practice.

None of the three external variables consistently reduced forecasting error. On average, median weekly earnings, adult-population growth, and the proportion aged 65 and over worsened validation performance by 0.4%, 0.6%, and 2.7%, respectively. They were therefore excluded from the test and forecasts.

Overall, these forecasts should be treated as conditional scenarios rather than policy targets. The short annual series and repeated cross-sectional survey design limit the strength of long-term interpretation, while recursive forecasting allows errors at earlier horizons to affect subsequent predictions. The ranges constructed from rolling-validation and 2022/23 test residuals are also indicative rather than formal prediction intervals. Borough-level and longer-term results should therefore be interpreted through their broad direction and relative differences rather than as precise point forecasts.

4.2 Evaluation of Physical Activity Forecasts by Age and Disability Group

4.2.1 Model Performance and Generalisation

Rolling-origin validation selected Gradient Boosting for four of the eight forecasting tasks. Naive persistence was selected for both age single-rate tasks, while Random Forest was selected for age activity composition and Ridge Regression for disability past-12-month participation. Independent testing in 2022/23 showed the largest gains over Naive for disability activity composition and disability overall composition, with improvements of 29.1% and 26.2% respectively. Age activity composition improved by approximately 15.4%, while the two age single-rate tasks did not improve on the Naive benchmark under the validation-selected specifications.

Table 4.1: Independent 2022/23 performance of the validation-selected RQ2 models. TV is used for compositional tasks and MAE for single-rate tasks.

Task	Selected model	Test error	Skill vs Naive
Age overall composition	Gradient Boosting	0.1083	21.4%
Disability overall composition	Gradient Boosting	0.1841	26.2%
Age activity composition	Random Forest	0.0080	15.4%
Disability activity composition	Gradient Boosting	0.0081	29.1%
Age past-12-month participation	Naive	0.0170	0.0%
Disability past-12-month participation	Ridge Regression	0.0212	5.5%
Age regular participation	Naive	0.0091	0.0%
Disability regular participation	Gradient Boosting	0.0092	15.5%

The clearest gains occurred in the composition and disability tasks. Age past-12-month participation remained well represented by persistence, while age regular participation was similarly best represented by the Naive benchmark. The supplementary mixed-effects comparison produced broadly similar component-level errors for the overall disability task, while Gradient Boosting remained the validation-selected model for that task.

4.2.2 Recursive Performance and Forecast Horizon

Historical recursive backtesting showed stable forecast behaviour across H1–H4. Age overall TV remained close to 0.11, changing from 0.1114 at H1 to 0.1115 at H4, while disability overall TV decreased from 0.1972 to 0.1810. Age activity-composition TV increased gradually from 0.0075 to 0.0093, and disability activity-composition TV remained close to 0.008 across the tested horizons (Figure 4.4).

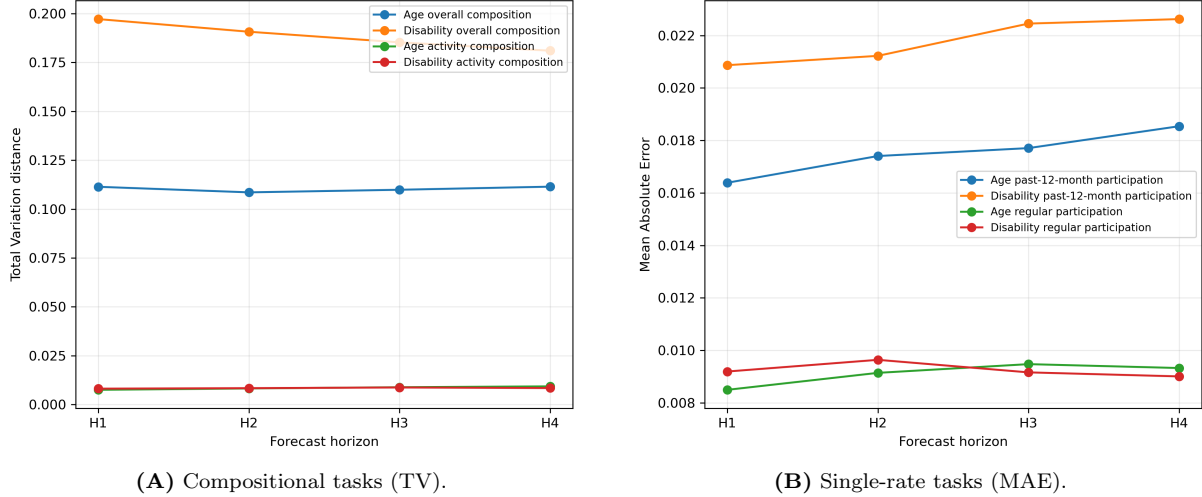


Figure 4.4: Historical recursive backtest performance from H1 to H4.

Age overall composition, disability overall composition, age activity composition, disability activity composition and disability regular participation outperformed recursive Naïve at all four tested horizons. Their mean gains were approximately 24.7%, 26.5%, 7.8%, 23.8% and 11.8%, respectively. Age past-12-month participation reproduced persistence, while the historical age regular-participation backtests showed smaller and less consistent gains across individual forecast origins.

Historical recursive evidence directly covers H1–H4. H4 is represented by only one historical forecast origin, while H5–H8 extend beyond the directly backtested range. The first four horizons therefore have stronger empirical support, while the later projections depend more heavily on the continuation of historical relationships and recursively generated inputs.

4.2.3 Forecast Patterns by Age, Disability and Activity

The production forecasts show relatively stable Active rates for most age groups, with a clear separation for the oldest groups. In 2023/24, several younger and middle-age groups had borough mean Active rates close to 0.70, compared with approximately 0.54 for ages 75–84 and 0.35 for ages 85+.

The 85+ group showed the largest movement between the final observed wave and the first forecast wave. Several borough-level observations were close to the boundaries of the proportion scale in 2022/23 and moved towards more moderate values in the first forecast year. These movements are consistent with model smoothing of highly variable cells with weak historical support.

The broad disability forecasts maintain a clear separation between the no-disability and limiting-disability groups. The limiting-disability group remains at a lower predicted Active rate throughout the selected-model forecast.

The 13 detailed disability indicators represent overlapping condition-specific groups and provide a more detailed view of projected participation patterns. Activity ranking was based on predicted past-12-month participation. Active Travel and Gardening appeared frequently among the highest-ranked activities across the age and disability strands. Age rankings remained stable because Naïve persistence was selected for the underlying age participation task, while the top-one to top-two margin indicates how clearly the leading activity is separated from the second-ranked activity.

4.2.4 Robustness and Limitations

Small-sample reliability was examined using Kish effective sample size. Supplementary shrinkage adjusted low-support borough predictions towards a leave-one-borough-out London reference while keeping the selected forecasting specification fixed. A primary threshold of $n_{\text{eff}} < 30$ was used to identify

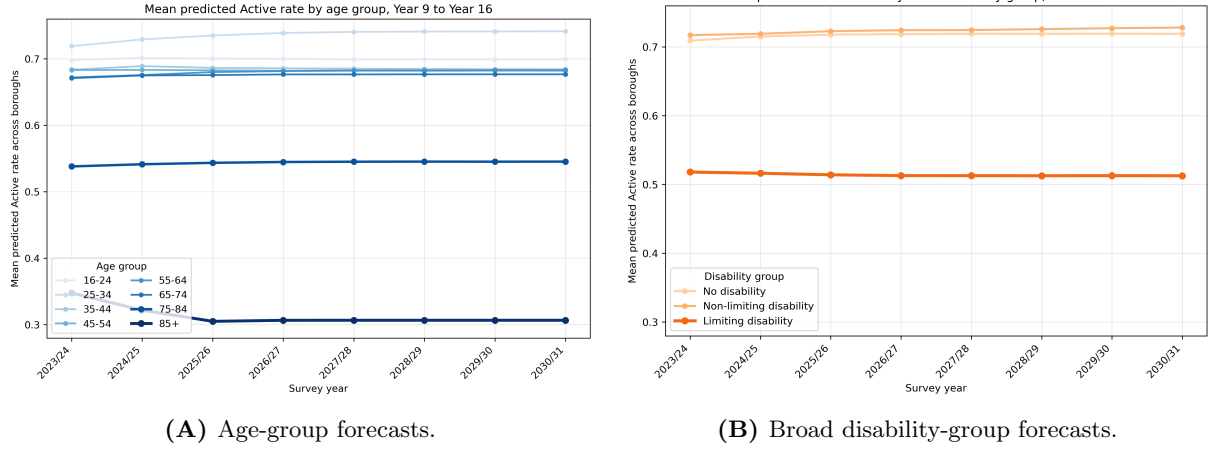


Figure 4.5: Mean predicted Active rate by age and disability group from 2023/24 to 2030/31. Panel (A) shows the forecasts for the eight age groups. Panel (B) shows the forecasts for the three broad disability groups.

low-support cells, with alternative thresholds included as sensitivity checks. The effect of shrinkage varied across forecasting tasks, so it remained a supplementary reliability analysis rather than part of the primary recursive forecast.

Training-weight sensitivity produced relatively small changes in the composition tasks. Replacing the main training weight with a rescaled square-root raw-sample-size weight changed age overall TV by approximately 0.9% and disability overall TV by -0.4% , while changes in the two activity-composition tasks remained below 3%. The disability single-rate tasks were more sensitive, with MAE changing by approximately 5.1% for past-12-month participation and 8.9% for regular participation.

The ALR zero-replacement sensitivity analysis compared $\epsilon = 10^{-4}$, 10^{-5} and 10^{-6} . The age overall TV result remained unchanged to the reported precision across the tested values, indicating low sensitivity to the numerical treatment of zero components.

A separate sensitivity analysis examined the COVID-period indicator. Table 4.2 reports the relative change in the primary test metric after removing the indicator.

Table 4.2: Change in primary test error after removing the COVID-period indicator. Positive values indicate higher error after removal.

Task	Change in primary test error
Age overall composition	-0.54%
Disability overall composition	$+0.08\%$
Age activity composition	$+1.51\%$
Disability activity composition	$+0.05\%$
Age past-12-month participation	0.00%
Disability past-12-month participation	-2.27%
Age regular participation	0.00%
Disability regular participation	$+0.28\%$

Most changes were small and varied in direction. The largest difference was a 2.27% reduction in MAE for disability past-12-month participation. Age past-12-month participation remained unchanged because the selected Naive specification does not use the COVID feature. Overall predictive performance was therefore only weakly sensitive to the COVID specification.

Residual-based recursive simulation generated 500 paths for the two overall composition tasks using held-out ALR residuals. The resulting 5th–95th percentile ranges are reported as indicative sensitivity bands (Appendix B.3). The disability strand showed a larger discrepancy between the deterministic forecast and the simulation median, indicating greater sensitivity to residual perturbation over the forecast horizon.

Cross-model comparison showed increasing divergence over the forecast horizon (Figure 4.6). For the 85+ group, Random Forest produced an unstable downward trajectory, with a sharp decline emerging in

the earlier part of the forecast period and Active rates settling at very low levels in the later years. The Random Forest trajectory is therefore used primarily as evidence of long-horizon model instability.

Among the remaining specifications, the main long-horizon difference was between Ridge Regression and Gradient Boosting. Ridge projected an increasing Active rate for the 85+ group, whereas Gradient Boosting projected an initial decline followed by a relatively stable trajectory. A similar difference appeared for the disability gap: Ridge projected substantial narrowing, while Gradient Boosting projected moderate widening. Random Forest produced a relatively stable widening trajectory for the disability gap, indicating that its instability was concentrated in the 85+ forecast rather than affecting both comparisons. The direction and magnitude of these longer-term changes are therefore more sensitive to model specification than the shorter-horizon forecast pattern.

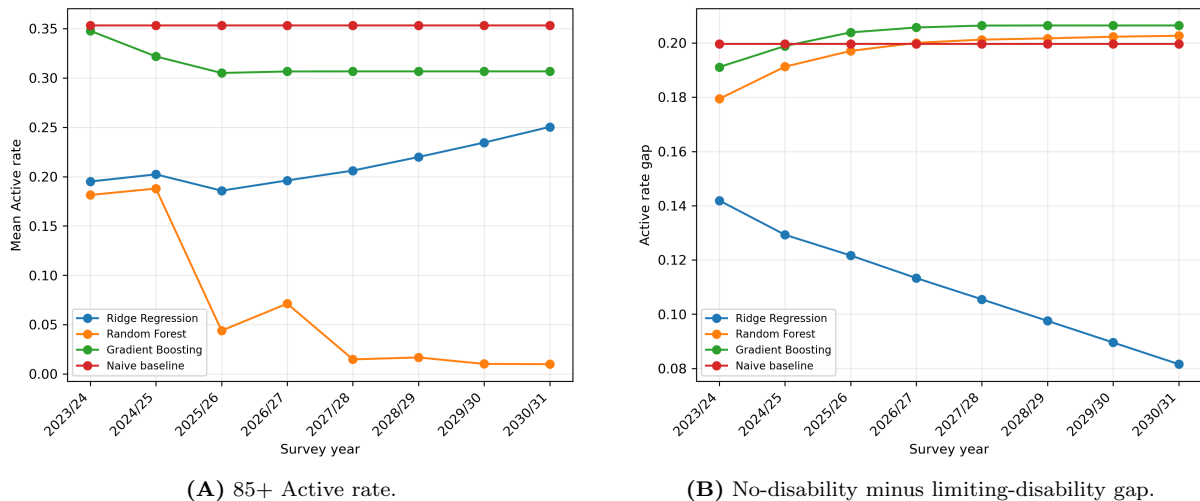


Figure 4.6: Cross-model long-horizon comparison from 2023/24 to 2030/31.

Historical recursive backtesting showed controlled forecast error over H1–H4. Cross-model comparison nevertheless showed increasing directional disagreement as the projection horizon lengthened. Long-horizon uncertainty is therefore concentrated mainly in the exact trajectory and rate of change, while recursive forecast errors remained controlled over the directly tested horizons.

Within the selected production forecasts, the oldest age groups remain at a substantially lower predicted Active rate than younger groups, and the limiting-disability group remains below the no-disability group. These relative demographic patterns are clear within the selected forecasting specification, while their exact long-term slopes remain more sensitive to model choice.

The evidence base is further limited by eight annual historical waves, low historical support in some borough-by-subgroup cells, and the repeated cross-sectional design of the Active Lives Survey. The forecasts therefore describe group-level participation patterns over time, with greater uncertainty for long-horizon and highly segmented estimates.

4.3 Evaluation of indoor/outdoor forecasts

Monthly and borough diagnostics supplement this evaluation in Appendix C.

4.3.1 Out-of-Sample Performance and Forecast Credibility

Three rolling folds validated on 2019/20, 2020/21 and 2021/22. The selected annual Random Forest used 250 trees, depth 8 and minimum leaf size 5; the monthly model used 250 trees, unrestricted depth and minimum leaf size 10. Selection preceded the 2022/23 test [10]. TV measures error across the four-state composition; MAE averages absolute errors in indoor and outdoor participation rates. Both use effective-sample-size weights within folds; CV averages the three fold scores equally.

In Table 4.3, annual and monthly Random Forest reduced test TV versus Naive by 18.20% and 17.60%, respectively, calculated as $100(1 - TV_{RF}/TV_{Naive})$. MAE 0.02300 means 2.30 percentage points, not a relative percentage error. These gains support one-year performance, not six-year accuracy; annual cells have stronger sample support.

Table 4.3: Three-fold validation means and held-out 2022/23 tests. Lower TV and MAE are better; skill measures TV reduction relative to Naive.

Frequency	Model	CV TV	CV MAE	Test TV	Test MAE	Test TV skill
Annual	Naive	0.04509	0.02553	0.04523	0.02842	0.00%
Annual	Random Forest	0.03648	0.02300	0.03700	0.02388	18.20%
Monthly	Naive	0.06564	0.04044	0.06319	0.03901	0.00%
Monthly	Random Forest	0.04982	0.03062	0.05207	0.03065	17.60%

4.3.2 Six-Year Balance and Activity-Level Differences

Historical London estimates aggregate partially pooled borough–activity-level compositions using survey weights. Indoor exposure share fell from 62.60% in 2017/18 to 57.39% in 2022/23. Mean annual shares were 61.75% across 2017/18–2019/20, 56.62% in 2020/21 and 56.51% across 2021/22–2022/23. The first period is not wholly pre-pandemic: 2019/20 covers November 2019–October 2020. Annual COVID controls starting in 2020/21 therefore approximate disruption coarsely. These changes complement pandemic research [11], without identifying a causal effect.

Forecasts hold borough–activity-level aggregation weights at their 2022/23 values. Under *persistent legacy*, indoor participation changes from 12.55% in 2023/24 to 12.28% in 2028/29, and outdoor participation from 9.38% to 9.45% (Figure 4.7). Indoor exposure share, calculated as indoor rate divided by the sum of both rates, changes from 57.23% to 56.52%. Under *legacy recovery*, it reaches 57.96%, a 1.44-percentage-point scenario difference. Both scenarios remain above 50% throughout the forecast horizon (Appendix Table C.3); this concerns recorded exposures, not the proportion of people participating indoors.

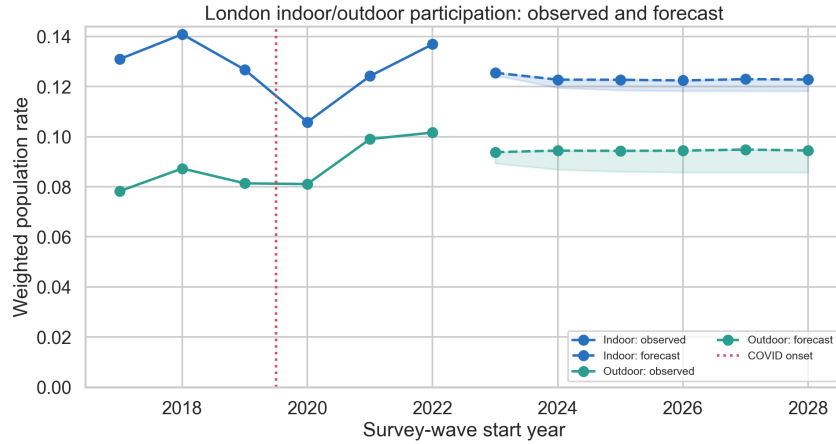


Figure 4.7: London participation rates, not exposure shares. Solid lines: pooled historical estimates; dashed lines: persistent legacy; shading: range between the two scenarios, not a prediction interval.

Among respondents with recorded indoor/outdoor exposure, Table 4.4 shows lower indoor exposure share for Active respondents than for either other group. These descriptive contrasts equalise total survey weight across waves; they are not regression-adjusted differences. A separate year- and borough-adjusted test rejected equal balances, $\chi^2(2) = 32.11$, $p < 0.001$. The Fairly Active–Inactive contrast was not significant, which does not establish equality. The evidence supports an Active-versus-other-groups distinction, not a demonstrated three-level gradient or a causal benefit of outdoor settings.

4.3.3 Limitations and Implications for London Sport

The predicted indoor exposure majority is conditional on these scenarios. Six waves cannot clearly separate trend, COVID disruption and borough interactions. Recursive forecasts propagate errors, and Random Forest extrapolates trends poorly. Scenario differences omit sampling and model-selection uncertainty. Moreover, 6,139 of 6,847 monthly cells have $n_{\text{eff}} < 20$; pooling stabilises estimates without

Table 4.4: London exposure-share contrasts across six waves, in percentage points. Intervals use 500 Bayesian-bootstrap draws and are not simultaneous. BH-FDR correction covers London and borough comparisons within this six-wave analysis.

Contrast (A–B)	Difference	95% bootstrap interval	FDR-adjusted p
Active–Inactive	–11.25	[–19.49, –2.44]	0.037
Active–Fairly Active	–9.12	[–12.33, –6.06]	0.014
Fairly Active–Inactive	–2.13	[–11.18, 6.63]	0.723

adding observations. Sparse Inactive exposure denominators also limit borough inference.

Missing settings were not imputed. The audited `neither_recorded` category separates no qualifying participation from qualifying participation without a recorded setting; neither implies physical inactivity. Respondents without recorded exposure remain in participation-rate estimates but cannot inform exposure-share contrasts. Survey weights do not remove recall error, non-response bias or unmeasured environmental influences.

For London Sport, these forecasts support accessible indoor provision while allowing for a narrower indoor advantage. Programmes for less-active residents should not assume the exposure balance observed among Active residents. Monthly and borough results should guide local validation; new survey waves should test issued forecasts before retraining.

4.4 Evaluation of Frequent Sport Volunteering Forecasts

4.4.1 Historical Patterns and Forecasting Context

The historical data from 2017/18 to 2022/23 provides a clear subgroup benchmark for subsequent forecast. Taking the median of the borough-median summaries across six survey years, the frequent sport volunteering rates for Active, Fairly Active and Inactive groups were 13.06%, 6.95% and 4.89%, respectively. When comparing by age group, the six-year summary of those aged 16–24 was the highest, at 14.26%. The Active-Inactive gap in history also varies by age group, while the Fairly Active-Inactive gap is generally close to zero in most age groups. These historical patterns constitute the reference for subsequent judgment on whether the forecast period continues similar subgroup structures.

The finer borough–age–activity_level panel contains a total of 4,608 theoretical cells, of which 4,290 have available volunteering rate observations. The median effective sample size of this panel was only 6.37. 81.5% of the observed cells were classified as small cells, and 39.8% of the observed cells had $n_{\text{eff}} < 5$. Therefore, historical fluctuations in the subdivision hierarchy may reflect the true subgroup heterogeneity, and may also be influenced by a large sampling variability. The activity-level differences observed here are descriptive associations, and it should not be assumed that the activity level will causally change the volunteering rate. For more details on historical trajectories, cell-support summaries and age-specific gaps, see Appendix D.

4.4.2 Predictive Performance and Robustness of Model Development

Table 4.5 summarizes the formal rolling validation results of three RQ4 forecasting tasks and the independent 2022/23 test performance. It shows that in rolling validation, the models selected are all superior to Naive baseline. The weighted MAE of borough–activity_level, borough–age and borough–age–activity_level decreased by 22.7%, 29.8% and respectively compared with Naive 29.3%. This advantage still exists in the independent 2022/23 test, with the corresponding error reduction magnitudes being 17.0%, 24.5%, and 23.5% respectively. Meanwhile, as the analysis hierarchy becomes finer, the prediction error also increases. This indicates that the borough–age–activity_level panels are more difficult to predict accurately compared to low-dimensional tasks.

The subgroup diagnostics in 2022/23 further indicates that an overall test error does not represent that all cells have the same predictive reliability. Within the borough–age–activity_level task, when $n_{\text{eff}} < 5$, the weighted MAE was 9.19 pp for cells, and dropped to 5.66 pp when $n_{\text{eff}} \geq 30$. The weighted MAE of the age 85+ group was 9.94 pp, and that of the Fairly Active group was 10.08 pp, both higher than the overall 6.83 pp of this task. Therefore, these results are more suitable for judging the overall prediction

Table 4.5: Rolling-validation and independent 2022/23 performance of the selected RQ4 models.

Task	Selected model	Rolling validation			2022/23 test		
		Weighted MAE <i>pp</i>	Naive MAE	Gain <i>%</i>	Weighted MAE <i>pp</i>	Naive MAE	Gain <i>%</i>
Borough-activity level	Random Forest	3.41	4.41	22.7	3.18	3.82	17.0
Borough-age	Gradient Boosting	4.71	6.71	29.8	5.01	6.64	24.5
Borough-age-activity level	Gradient Boosting	6.25	8.84	29.3	6.83	8.92	23.5

Note: MAE denotes effective- n -weighted mean absolute error. It is numerically identical to the weighted Total Variation. Rolling-validation values are means across the 2019/20–2021/22 validation periods.

differences of aggregate subgroup patterns, but it cannot be assumed that the predictions of all cells are equally reliable. Full cell-support diagnostics are provided in Appendix D.

Model development sensitivity checks are used to determine whether the above model selection overly relies on a specific evaluation design. First, after re-ranking the models using unweighted MAE on exactly the same rolling validation predictions, the original formal model specification selection was retained for all three tasks. It indicates that model selection is not solely determined by effective- n weighting.

The COVID-related checks produced slightly weaker evidence for the exact candidate ranking. All three formally selected models still outperformed Naive in the independent 2022/23 test. Further, after not including 2019/20 when calculating validation scores, borough-activity_level retained Random Forest. The optimal model of borough-age changed from Gradient Boosting to Ridge Regression, but the weighted MAE of the two differed by only about 0.03 pp. The borough-age-activity_level task still selected Gradient Boosting, but the specific parameter combination changed. The weighted MAE of the two specifications only differs by approximately 0.05 pp. Thus the precise ranking is somewhat sensitive to the COVID period, but the differences in the main prediction errors corresponding to these changes are very small. Furthermore, due to the simultaneous changes in volunteering measurement and survey weighting in 2019/20, these checks can only be used to assess the robustness of model development and cannot identify the causal effect of COVID-19 on volunteering.

4.4.3 Forecast Findings and Historical Consistency

Figure 4.8 presents the predicted changes from 2023/24 to 2028/29 based on two forecasting tasks, borough-age and borough-activity_level. Each line represents the median projected rate of the corresponding subgroup in 32 London boroughs. Panel (A) shows that the borough-median forecasts for the eight age groups remained unchanged from 2023/24 to 2028/29. The predicted values of the four groups aged 25–34, 35–44, 55–64 and 65–74 completely overlap, and the borough median of the six prediction periods is approximately 10.21%. The 16–24 group has the highest borough-median projection, at approximately 14.69%, significantly higher than that of other age groups. The 85+ group is the lowest, only approximately 3.09%. These flat and overlapping trajectories are a model-generated recursive forecast pattern generated by borough-age Gradient Boosting, rather than a directly observed future change. Therefore, it cannot be assumed that the real volunteering rates of these age groups will remain unchanged in the next six years.

Panel (B) shows a clearer activity-level ordering. The borough-median rate of the Active group is consistently higher than that of the Fairly Active and Inactive groups during the forecast periods. The three trajectories show little long-term divergence between 2023/24 and 2028/29, with the main changes concentrated in the earlier forecast periods. Therefore, compared with the historical benchmarks in Section 4.4.1, these predictions retain the overall ranking of activity levels ordering and the differences between the youngest and oldest age groups, although the exact long-term levels still depend on the fitted models and recursive assumptions.

Figure 4.9 compares the activity-level volunteering gap of different age groups in historical period and forecast period. Panel (A) is Active-Inactive gap, and Panel (B) is Fairly Active-Inactive gap. For each point, the borough-level gap is first obtained by subtracting the frequent-volunteering rates of two types of activity levels within the same borough, age group, and survey year. Then take the median of boroughs, and further take the median of the annual medians for the six periods from 2017/18 to 2022/23 or 2023/24 to 2028/29.

Panel (A) shows that the Active-Inactive gap in the forecast period is positive in all age groups and continues to exhibit significant age dependence. The median gap of forecast for those aged 16–24 and 45–54 is the largest, approximately 7.57 pp and 7.27 pp, while for those aged 25–34, it is only about 2.95 pp. Compared with the historical profile, the uneven structure among different age groups still exists, but the

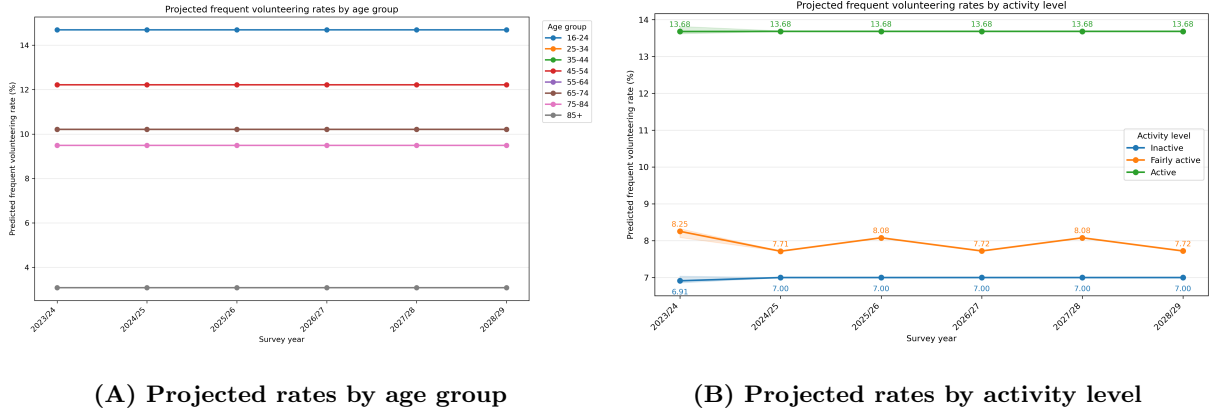


Figure 4.8: Projected frequent sport volunteering rates from 2023/24 to 2028/29.

gap between the ages of 16–84 has generally narrowed. The forecast period basically follows the overall pattern of gap changes with age in the historical period, but the difference range is generally weaker than that in the historical period.

The pattern of Panel (B) is obviously different. Historical Fairly active-Inactive gaps are close to zero in all age groups, and the 25–64 age group in forecast also basically maintains this structure. The more obvious deviations mainly occur at both ends of age, especially for those over 85, the forecast gap is approximately 4.2 pp. However, 85+ belongs to the subgroup with relatively weak sample support and relatively high 2022/23 test error. Therefore, this outstanding result should not be regarded as an equally reliable age difference.

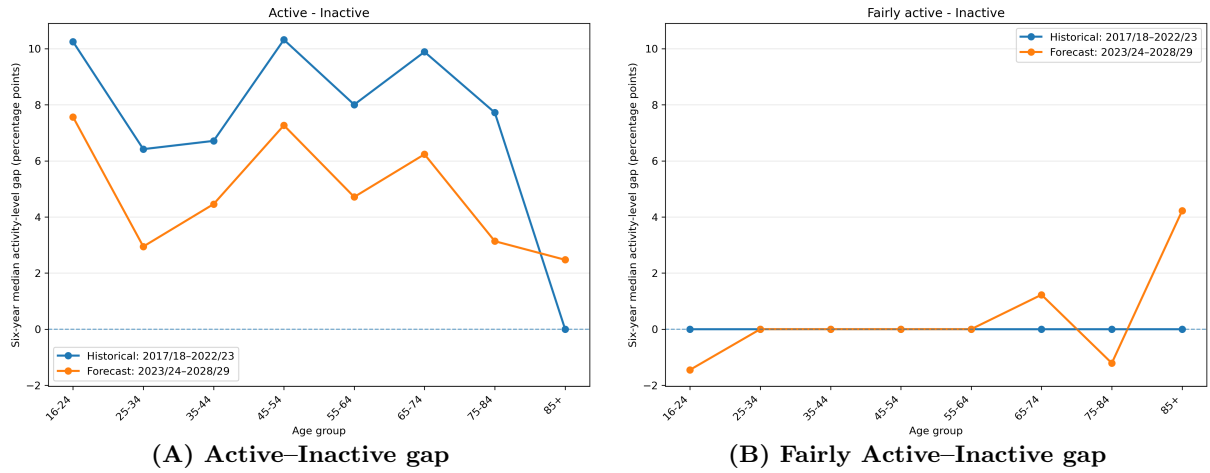


Figure 4.9: Historical and forecast age-specific activity-level gaps.

4.4.4 Forecast Robustness and Strength of Evidence

Recursive backtesting provides direct evidence for one to four-step forecasting. Mean effective- n -weighted MAE for borough-activity_level task ranged from 2.75 to 3.40 pp, for borough-age is 4.63–5.26 pp, and for borough-age-activity_level is 6.14–6.99 pp. Therefore, within the directly verifiable range, there is no persistent and rapid error accumulation. Since historical data can directly test at most four recursive forecasts, while formal predictions extend to six steps, there are no corresponding historical backtests for the predictions of 2027/28 and 2028/29, empirical support is consequently strongest for the earlier forecast years.

The Structural Model-family sensitivity further examines the dependence of the predictions of the three forecasting tasks on the model form. Table 4.6 compared the alternative model and primary model of each task cell by cell in 2023/24 and 2028/29 prediction differences. The prediction of borough-activity_level task is the most stable overall. Borough-age showed relatively small differences in most cells, but the P90 difference of Ridge Regression exceeded 7 pp in both 2023/24 and 2028/29, thus part

of borough-age cells are relatively sensitive to model form. This sensitivity is more pronounced in the finest granularity of borough-age-activity_level task. In 2023/24, the median differences between Random Forest and Ridge Regression relative to primary Gradient Boosting are 2.11 and 3.69 pp respectively. By 2028/29, the figures are 1.18 and 2.41 pp respectively.

But these differences did not expand consistently with forecast horizon. Therefore, it is not the case that the farther the prediction, the stronger the model-form sensitivity. This check indicates that the overall structure of primary forecasts is stable, but the specific predicted values of some subgroups and sub-cells still significantly depend on the adopted model family. Since what is compared here is the prediction differences between models rather than the prediction error, it is used to assess the robustness of the forecast rather than to reselect the primary model. Detailed recursive diagnostics and gap-specific model-family comparisons are reported in Appendix D.

Table 4.6: Structural model-family sensitivity of RQ4 projections.

Task (primary)	Alternative	2023/24		2028/29	
		Median	P90	Median	P90
Borough-activity (RF)	Gradient Boosting	0.34	1.20	0.26	0.55
	Ridge Regression	0.98	1.89	3.10	3.62
Borough-age (GB)	Random Forest	1.14	3.13	0.79	4.17
	Ridge Regression	1.10	7.45	1.26	7.75
Borough-age-activity (GB)	Random Forest	2.11	4.30	1.18	3.17
	Ridge Regression	3.69	5.75	2.41	4.82

*Note:*Median and P90 denote the median and 90th percentile of the cell-level absolute difference between each alternative model and the task specific primary model.

Chapter 5

Conclusion

This study focuses on the practical needs of London Sport regarding future physical activity planning and, based on the Active Lives Adult Survey, has established a coordinated data analysis and forecasting framework covering geographical areas, population groups, activity settings and sport volunteering. The four research questions examine geographical and temporal patterns across London overall, Inner/Outer London and the boroughs; age and disability groups; indoor/outdoor participation; and frequent sport volunteering. Overall, this study has achieved the four analytical and forecasting objectives set out, and has provided London Sport with forward-looking evidence that can be used for borough prioritisation, population-group targeting, activity provision and volunteering planning. However, given the short historical time series, the sparsity of some subgroup cells, the fact that Active Lives is a repeated cross-sectional survey, and the increasing reliance of long-horizon forecasts on model-generated values, all future results should be understood as conditional, model-based projections rather than deterministic forecasts or causal estimates.

5.1 Main Contributions and Achievements

RQ1: Physical Activity Trends, Borough Differences and Forecasts. RQ1 constructed a multi-scale analysis framework covering London as a whole, inner London, outer London, and 32 boroughs, and evaluated physical activity patterns by combining temporal trends, spatial differences, clustering, and predictive analysis. From 2015/16 to 2022/23, the proportion of Active in London increased by 1.8 percentage points, the Inactive proportion also increased by 1.6 percentage points, while the Fairly Active proportion decreased by 3.4 percentage points, indicating that the composition of physical activity was more consistent with redistribution towards two extremes rather than an overall improvement. The average Active proportion in inner London was 5.6 percentage points higher than that in outer London, and the gap remained 4.5 percentage points after age standardisation. The relative differences among boroughs also showed clear persistence and spatial clustering. In the formal annual predictions, the Naive model was selected for London as a whole and inner/outer London, while Ridge Regression was selected at the borough level; the weighted TV errors for the independent test in 2022/23 were 0.66, 0.70, and 3.99 percentage points, respectively. Therefore, the main annual predictions did not support a clear long-term direction of change for London as a whole or inner/outer London. As a supplementary scenario, the annualised quarterly predictions showed that the Active proportion in London might increase by approximately 2.0 percentage points by 2030/31, but the changes in inner London and outer London were smaller and inconsistent in direction. The median change in the Active proportion in borough-level predictions was approximately +1.1 percentage points, but the final levels showed clear shrinkage. Therefore, the convergence in predictions cannot be interpreted as the inevitable narrowing of borough-level gaps in reality. Overall, RQ1 identified persistent geographical differences and provided conditional scenarios that can be used for planning, but the existing data are insufficient to determine a unified and clear long-term trend across all areas.

RQ2: Physical Activity Forecasts by Age and Disability Group. RQ2 developed eight forecasting tasks covering overall activity composition, activity-specific composition, past-12-month participation and regular participation across age and disability groups. Rolling-origin validation selected Gradient Boosting for four of the eight tasks. Naive persistence was selected for both age single-rate tasks, while

Random Forest was selected for age activity composition and Ridge Regression for disability past-12-month participation. On the independent 2022/23 test wave, the largest improvements over Naive were observed for disability activity composition and disability overall composition, at 29.1% and 26.2% respectively. Historical recursive backtesting showed controlled forecast error across H1–H4, providing stronger empirical support for the earlier forecast horizons. In the production forecasts, the oldest age groups remained at substantially lower predicted Active rates than younger groups, while the limiting-disability group remained below the no-disability group throughout the forecast period. Active Travel and Gardening appeared frequently among the activities with the highest predicted participation rates. Cross-model comparison showed that the precise long-term direction of the 85+ trajectory and the disability gap was sensitive to model specification, while the selected production forecasts continued to show clear differences between demographic groups. Overall, RQ2 provides age- and disability-specific forecasting evidence within the limits of the available eight-year history.

RQ3: Indoor and Outdoor Participation Forecasting RQ3 harmonised the six 2017/18–2022/23 waves and retained 146 of 179 activities with complete Months, Days, indoor and outdoor field families. Qualifying participation required *Months* = 1 and *Days* ∈ {1, 2}. The four-part model retained `neither_recorded`, `indoor_only`, `outdoor_only` and `both`, while the audit separated `neither_recorded` into no qualifying participation and qualifying participation without a recorded setting. Three rolling-origin folds selected Random Forest for both frequencies. On the untouched 2022/23 test, annual TV/-MAE were 0.03700/0.02388 and monthly TV/MAE were 0.05207/0.03065, improving on Naive TV by 18.20% and 17.60%. Under persistent legacy, indoor exposure changes from 57.23% in 2023/24 to 56.52% in 2028/29; legacy recovery gives 57.96% in 2028/29. Active respondents had significantly lower observed indoor exposure than Inactive and Fairly Active respondents, although the latter two did not differ significantly. RQ3 therefore achieved its London-wide annual, monthly, activity-level and borough forecasting objectives, but supports conditional exposure-balance scenarios rather than causal effects, precise borough targets or calibrated long-horizon intervals.

RQ4: Frequent Sport Volunteering Forecasts RQ4 constructed three complementary borough-level forecasting tasks for frequent sport volunteering: `borough-activity_level`, `borough-age` and `borough-age-activity_level`. The selected task specific models, in the independent 2022/23 test, reduced the effective-*n*-weighted MAE by 17.0–24.5% compared with the Naive baseline. The predictions for 2023/24–2028/29 generally continue the historical subgroup structure: The activity-level central forecast of the Active group was the highest, the age specific borough-median prediction of the 16–24 age group was the highest, and the 85+ group was the lowest. The predicted activity-level gap varies with age, and this difference is mainly reflected in the Active-Inactive contrast. Its age-specific pattern is generally similar to the historical period. However, for most age groups aged 16–84, the gap of the forecast period is generally smaller than that of the historical period. By contrast, the Fairly Active-Inactive gap remains close to zero in most age groups. Overall, the prediction of RQ4 identified subgroup-level patterns and their relative differences, rather than the precise long-term rate of a certain cell. Recursive backtesting shows that within the directly testable H1–H4 horizons, the prediction error does not continue to deteriorate as the number of horizon increases. However, comparisons among different model families indicate that some specific cell-level forecast values can be significantly influenced by model choice. Therefore, these results are only conditional forecasts under the condition of the continuation of historical relations.

5.2 Current Limits and Further Work

These four research questions share a number of common limitations. The most significant of these are the short length of historical data series, the scarcity or low reliability of sub-group data points, and the uncertainty arising from long-term recursive forecasting. Longer-term forecasts rely increasingly on relationships identified during the observation period, as well as lag data generated by the model. In addition, the forecast ranges and sensitivity analyses used in this study are intended as sensitivity indicators rather than calibrated prediction intervals.

For RQ1, Active Lives is a repeated cross-sectional survey, so the analysis captures population-level changes across survey waves rather than longitudinal behavioural change among the same individuals, while more detailed geographical estimates are subject to greater sampling variability. For RQ2, historical support is weaker for the oldest age groups and the detailed disability indicators, and cross-model comparison showed that the long-term direction of the 85+ Active-rate trajectory and the disability gap

varied across model specifications. Activity rankings also remain uncertain when the two leading predicted participation rates are close. For RQ3, 6,139 of the 6,847 monthly cells had an effective sample size below 20; partial pooling stabilises these estimates but cannot create information absent from the underlying data. Random Forest also has limited ability to extrapolate trends beyond its observed training range. The COVID-19 specifications represent conditional sensitivity scenarios rather than estimates of a causal pandemic effect, and incomplete setting data mean that indoor exposure must be interpreted as the balance of recorded indoor and outdoor exposures rather than the proportion of people who participated indoors. For RQ4, the cell-level sample support of the borough-age-activity_level panel is the weakest, and the diagnostic results of 2022/23 show that predictive reliability varies across effective-sample-size bands and subgroups. The six-year recursive forecast also exceeds the four-step prediction range that can be directly verified by the current historical data. Meanwhile, the comparison of different model families indicates that some cell-level projections are affected by model choice. Furthermore, the cell-level and borough-level summaries reported in this paper are not London-wide forecasts obtained through population weighting.

Future work should prioritise prospective validation as new Active Lives survey waves become available. Existing forecasts should be evaluated against new observations before models are retrained and updated, while a longer historical series would provide more rolling-origin forecast points and allow stronger assessment of long-horizon stability. Multilevel, hierarchical or Bayesian approaches could also be explored for sparse borough and subgroup estimates, and consistently measured external predictors such as facility accessibility, population structure, transport, income, deprivation, weather and green-space provision could be examined where reliable historical and future values are available. For RQ2, activity rankings should also be updated as new waves are added; for RQ3, more complete indoor/outdoor setting fields and calibrated prediction intervals would further improve the forecasts. For RQ4, should corresponding population projections become available in the future, it would also be possible to derive population-weighted estimates at the London level.

5.3 Recommendations for London Sport

The research results support that London Sport should adopt a more targeted and adaptable planning approach. Resource decisions should consider geographical patterns, subgroup needs, accessibility of activity opportunities, and sport volunteering provision, while recognising that forecasts are conditional scenarios requiring local assessment and future validation rather than fixed predictions or causal conclusions.

First, London Sport should monitor changes across all three physical activity levels rather than focusing only on the proportion of residents classified as Active. The supplementary annualised quarterly scenario suggests that the Active proportion may increase, while Fairly Active may continue declining and Inactive may rise slightly. Early intervention should therefore focus on helping Fairly Active residents sustain participation and supporting Inactive residents to access opportunities.

Second, more support should be directed towards boroughs with persistently lower activity levels. Newham and Barking and Dagenham were identified as Low-Low spatial clusters and could be prioritised for local consultation, co-designed interventions, and pilot programmes. Higher-activity areas such as Wandsworth could be used as comparators to explore which local conditions or practices may be transferable. Resource allocation should also consider cost, transport access, facility opening hours, venue quality, and local needs so that interventions are appropriate to each borough.

Third, the annual model selection results should be used as the main basis for medium-term planning. Annualised quarterly predictions can serve as supplementary scenarios to reflect seasonality and possible changes, but they should not replace the formal annual predictions. Due to the greater sampling fluctuations and longer recursive forecasting horizons of monthly predictions, they should only be regarded as exploratory evidence. When new Active Lives data are released, the new observations should first be used to conduct prospective validation of the existing predictions, after which the models should be retrained and future predictions updated.

Fourth, London Sport should provide additional support tailored to older adults and disabled people. The forecasts suggest that the activity gap between people with limiting disabilities and those without disabilities may widen, while people aged 85 and over are expected to remain the least-active age group. London Sport could therefore expand affordable, accessible, locally available, and flexibly scheduled opportunities, including community-based indoor sessions, walking programmes, and gardening activities, to reduce barriers to participation.

Fifth, London Sport should expand outdoor opportunities while maintaining accessible indoor provision. By 2028/29, both scenarios retain an indoor majority in recorded exposure balance, while less-active groups have higher observed indoor exposure shares than Active respondents. This measure is not the proportion of residents participating indoors. London Sport could continue supporting accessible indoor opportunities while piloting links to park walking, cycling, or outdoor programmes and evaluating their accessibility and effects on participation.

Sixth, sport volunteering could be integrated more naturally into participation programmes. Active respondents generally have higher rates of frequent sport volunteering, with particularly large modelled Active–Inactive gaps among those aged 16–24, 45–54, and 65–74. London Sport could offer accessible, low-commitment volunteering opportunities for these age groups, such as assisting with community sessions, supporting events, or taking on peer-support roles. Pilot programmes could then assess whether these opportunities contribute to longer-term participation and volunteer retention.

Overall, London Sport should prioritise pilot programmes in boroughs with greater identified need and evaluate changes in the Inactive proportion, continued participation among the Fairly Active group, reach among older adults and disabled residents, use of indoor and outdoor opportunities, and volunteer recruitment and retention. This approach would help direct limited resources more effectively towards reducing persistent geographical and population-group inequalities.

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Appendix A

Supplementary Material for Activity Trends, Boroughs and Forecasting

This appendix presents supplementary temporal, spatial, clustering, and forecasting results for RQ1.

A.1 Supplementary Temporal and Spatial Patterns

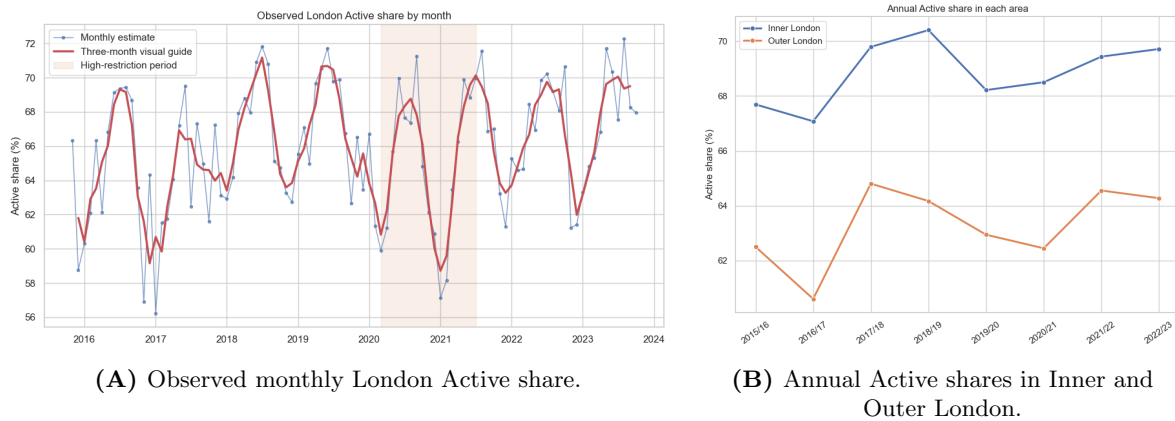


Figure A.1: Supplementary observed temporal patterns. Panel (A) shows monthly variation, while Panel (B) compares annual Inner and Outer London estimates.

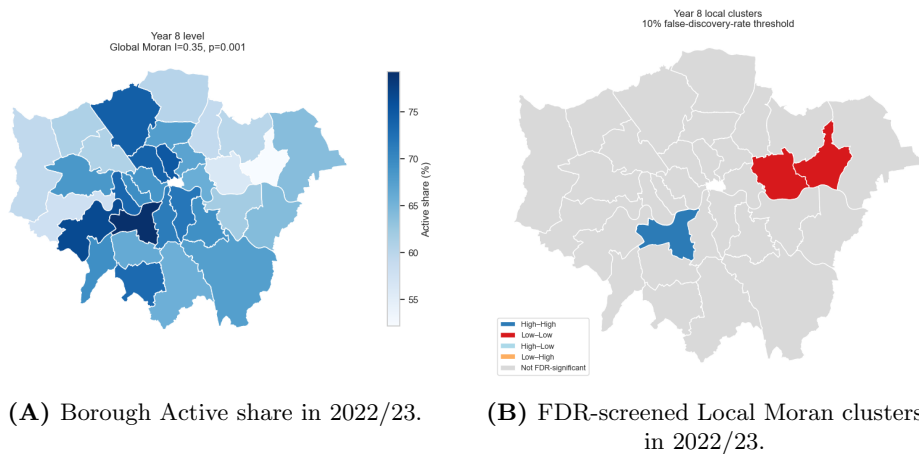


Figure A.2: Observed borough spatial patterns. Panel (A) presents Active levels, and Panel (B) identifies significant local clusters.

Contextual data sources and licensing. The contextual datasets used in RQ1 were obtained from publicly available sources. The ONS population estimates, population change data, and Annual Survey of Hours and Earnings data are provided under the Open Government Licence v3.0, while the borough-level Metropolitan Police crime data were accessed through the London Datastore under the Open Government Licence v2. All datasets were used solely for academic analysis and cited in accordance with the relevant providers' attribution requirements [42, 43, 44].

A.2 Structural Clustering Diagnostics

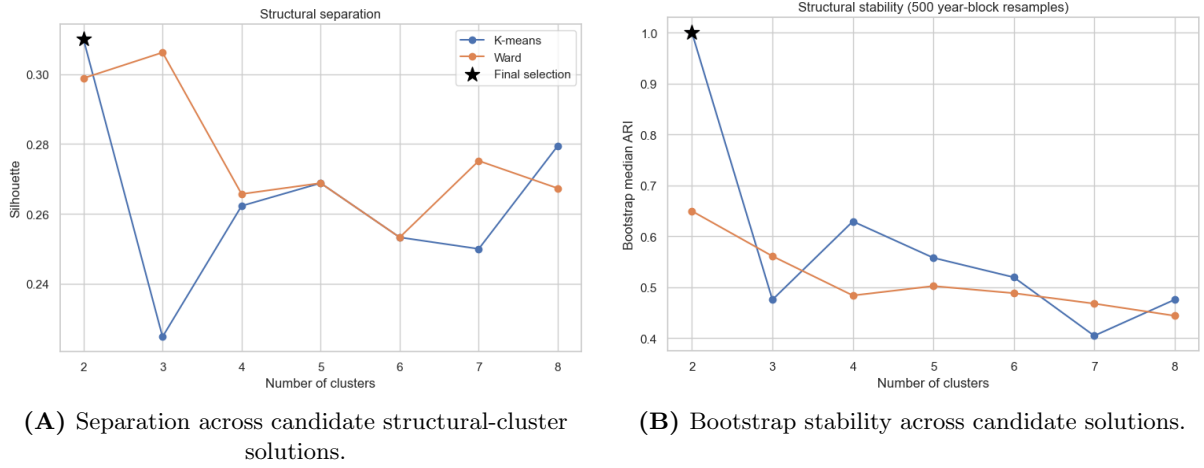


Figure A.3: Selection diagnostics for the structural typology. Stars identify the selected two-cluster K-means solution.

A.3 Supplementary Forecast Results

The London monthly Ridge forecast retains a clear seasonal cycle, with modest increases in Active and Inactive and a gradual decline in Fairly Active over the forecast horizon (Figure A.4B). However, its untouched 2022/23 weighted TV error was 2.25 percentage points, compared with 1.66 for the quarterly model and 0.66 for the annual model (Figure A.4A). The widening sensitivity bands and 96-step recursive horizon further limit long-term precision, so the monthly results are treated as supplementary evidence rather than the primary forecast.

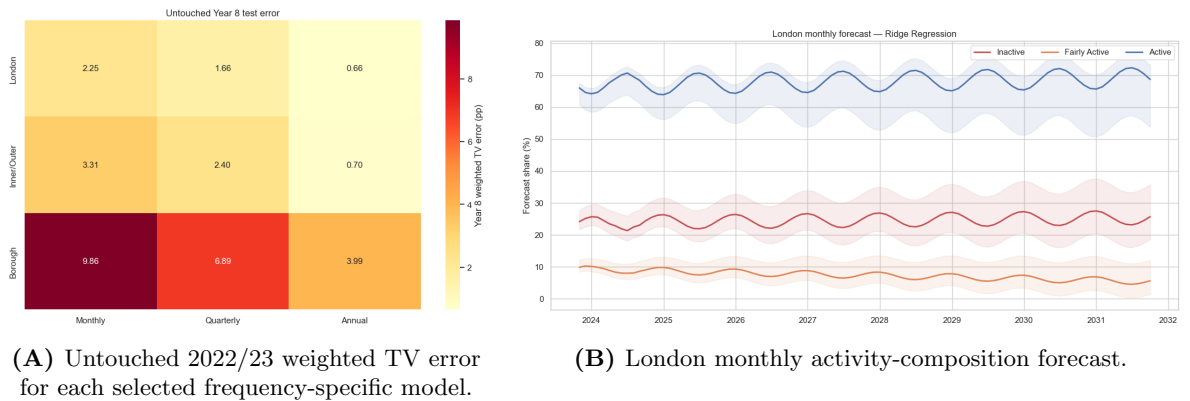


Figure A.4: Supplementary forecast evaluation and monthly results. Panel (A) compares test errors across geographical and temporal resolutions; Panel (B) presents the London monthly forecast.

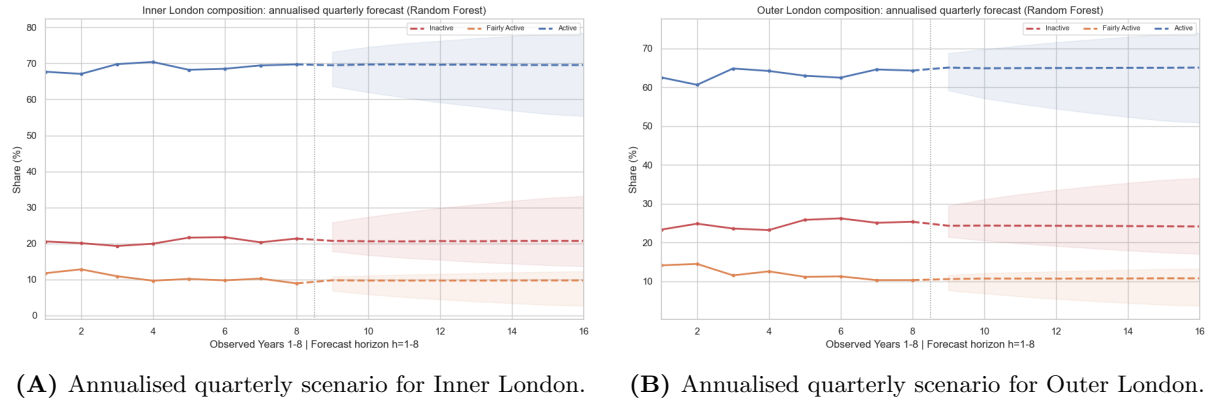


Figure A.5: Supplementary annualised quarterly scenarios for Inner and Outer London. Shaded ranges are sensitivity bands rather than formal prediction intervals.

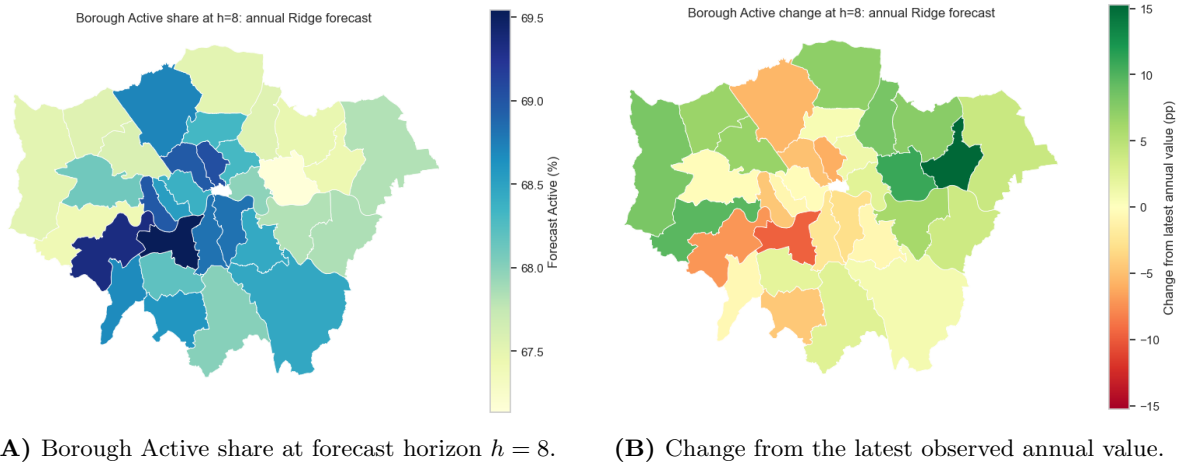


Figure A.6: Borough annual Ridge forecasts at $h = 8$. Panel (A) shows final forecast levels, while Panel (B) shows changes from the latest observed values.

Appendix B

Supplementary Material for Age and Disability Forecasting

B.1 Candidate-Model Comparison

Figure B.1 shows the candidate-model comparison across the eight RQ2 forecasting tasks. The figure reports the task-specific primary validation metric used during model selection. Lower values indicate better validation performance. Model family and hyperparameters were selected from rolling-origin validation only, while the 2022/23 test wave remained outside the selection process.

B.2 Sample-Support Diagnostics

Historical sample support varied across the age analysis after the data were divided by borough and age group. Figure B.2 compares Kish effective sample size with absolute prediction error for the age activity-level task on the independent 2022/23 test wave.

B.3 Residual-Based Recursive Simulation

Residual-based recursive simulation was used as a supplementary sensitivity analysis for the age and disability overall-composition forecasts. Five hundred forecast paths were generated using held-out ALR residuals and propagated recursively across the eight-year forecast horizon. The 5th–95th percentile ranges are presented as indicative sensitivity bands.

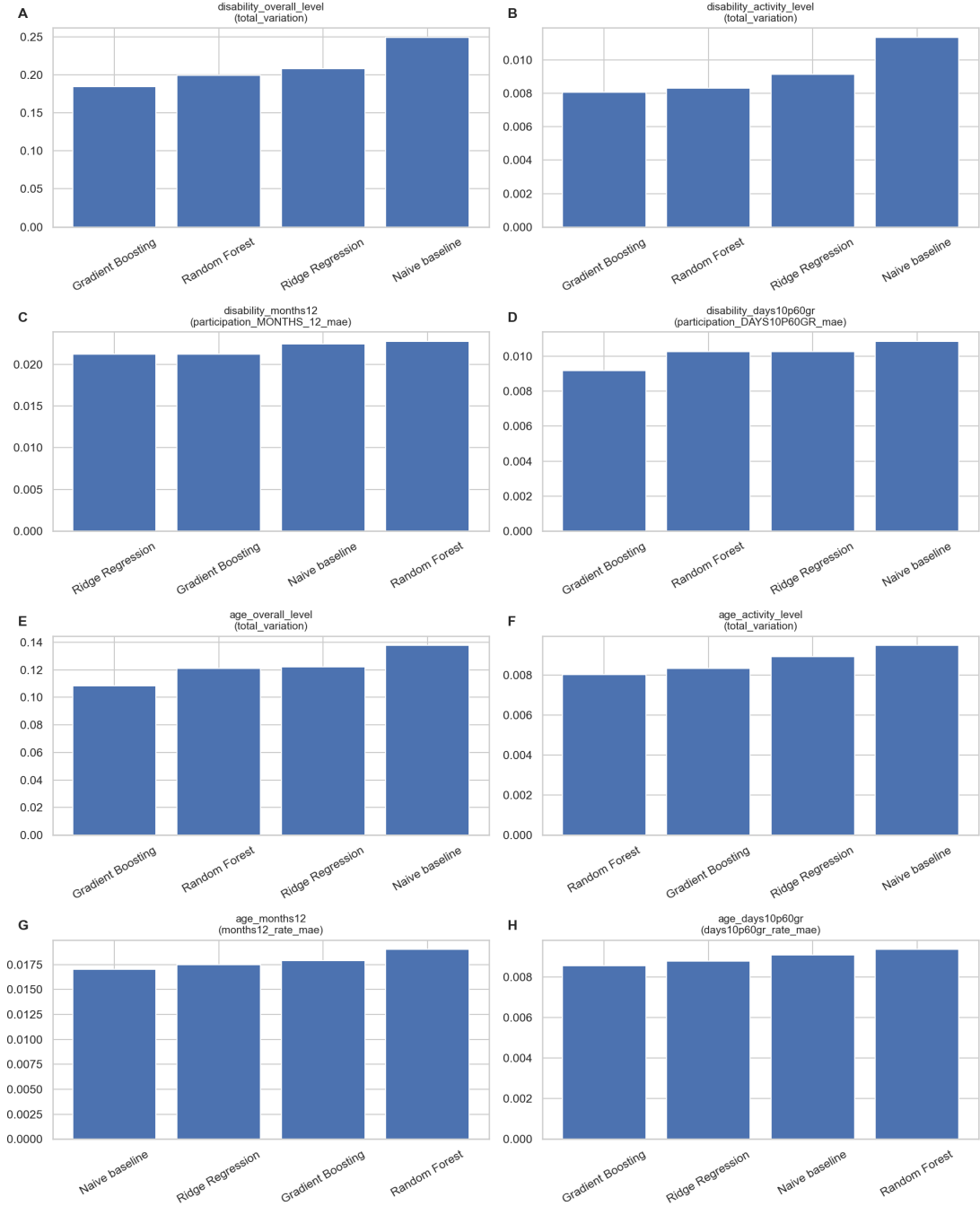


Figure B.1: Candidate-model comparison across the eight RQ2 forecasting tasks. Total Variation is used for compositional tasks and MAE for single-rate tasks. Lower values indicate better rolling-origin validation performance.

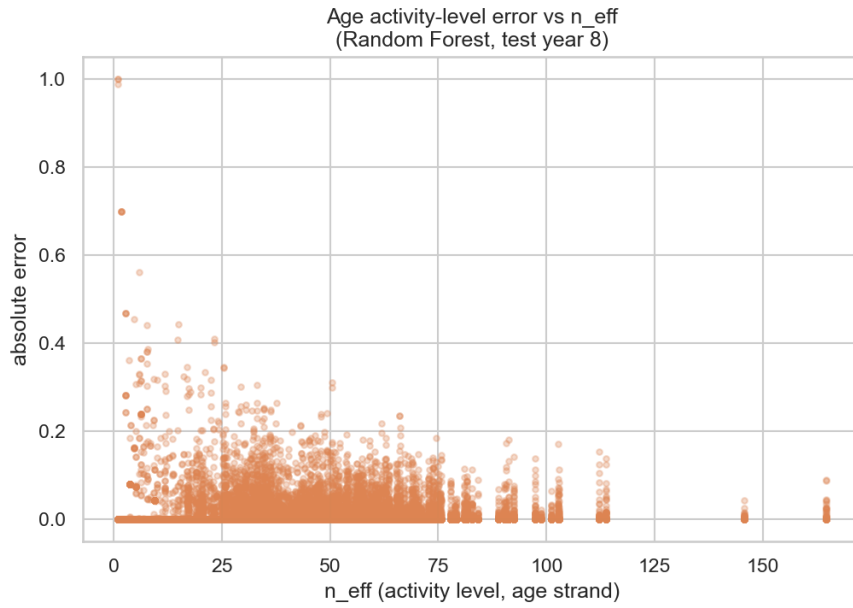


Figure B.2: Absolute prediction error against Kish effective sample size for the age activity-level task on the independent 2022/23 test wave.

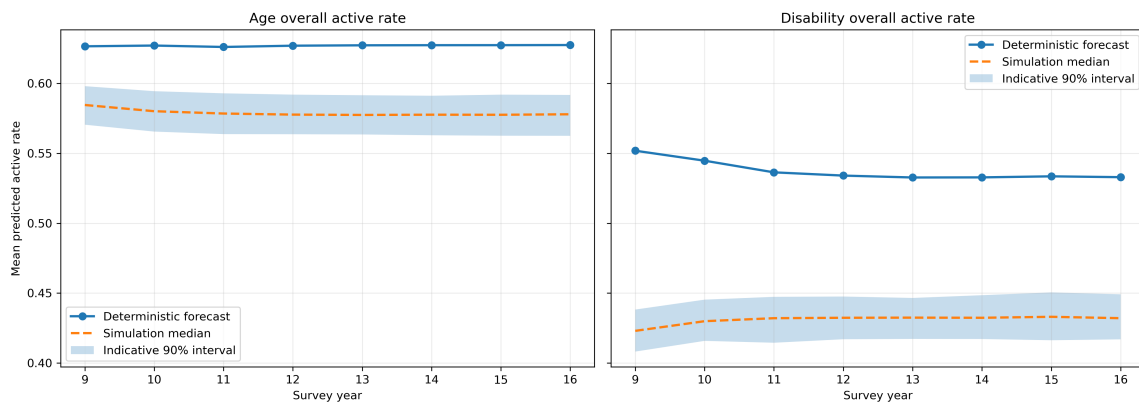


Figure B.3: Residual-based recursive simulation for the age and disability overall-composition forecasts. The solid line shows the deterministic forecast, the dashed line shows the simulation median, and the shaded area represents the 5th–95th percentile indicative sensitivity band across 500 simulated paths.

Appendix C

Supplementary Material for Indoor/Outdoor Forecasting

This appendix preserves the detailed algorithm, parameter audit, complete annual forecasts and monthly/borough diagnostics supporting Sections 3.4 and 4.3. The algorithm is intentionally omitted from a front-matter List of Algorithms, and no code listing is included.

C.1 Outcome Construction, Validation and Candidate Models

Input: Respondent i 's Months, Days, indoor, and outdoor values for 146 comparable activities

Output: One four-part state and two reconciled `neither_recorded` subcategories

$I_i \leftarrow 0, O_i \leftarrow 0, P_i \leftarrow 0$

foreach comparable activity j **do**

$Q_{ij} \leftarrow (Months_{ij} = 1) \wedge (Days_{ij} \in \{1, 2\})$

$P_i \leftarrow P_i \vee Q_{ij}$

$I_i \leftarrow I_i \vee (Q_{ij} \wedge Indoor_{ij} = 1)$

$O_i \leftarrow O_i \vee (Q_{ij} \wedge Outdoor_{ij} = 1)$

end

Assign `neither_recorded`, `indoor_only`, `outdoor_only`, or `both` from (I_i, O_i)

Within `neither_recorded`, flag $P_i = 0$ as no qualifying activity

Within `neither_recorded`, flag $P_i = 1$ as qualifying activity without a recorded setting

Assert that the two flags sum exactly to `neither_recorded`

Algorithm C.1: Construction and reconciliation of respondent-level RQ3 setting states

Table C.1 demonstrates that the two meanings previously hidden by “neither” are stored separately. Neither subcategory is inferred from `MEMS7GR_ALL`; consequently, `neither_recorded` must not be described as the survey’s Inactive group.

Table C.1: Reconciliation of the two `neither_recorded` subcategories by observed wave.

Wave	No qualifying activity	Activity without setting	Total neither recorded
2017/18	7,438	5,983	13,421
2018/19	7,492	5,683	13,175
2019/20	8,478	5,058	13,536
2020/21	9,161	4,541	13,702
2021/22	8,389	5,072	13,461
2022/23	8,139	5,457	13,596

The explicitly dated validation targets are 2019/20, 2020/21 and 2021/22; 2022/23 is excluded from every choice. The 250-tree budget balances stable forest averaging with reproducible runtime, while the two depth/leaf settings compare moderate and flexible trees. Ridge penalties and boosting rates/depths

Table C.2: Candidate models and intensity parameters evaluated in each rolling-origin fold.

Model	Candidate settings
Naive	Copy the previous wave or corresponding previous-year month
Ridge	$\alpha \in \{0.1, 1, 10, 100\}$
Random Forest	250 trees; $(max_depth, min_samples_leaf) \in \{(8, 5), (None, 10)\}$
Gradient Boosting	$(n, lr, depth, leaf) \in \{(120, 0.03, 2, 5), (180, 0.03, 2, 10), (120, 0.05, 3, 10)\}$

span weaker and stronger regularisation, with all intensity parameters decided by joint mean TV/MAE rank.

C.2 Complete Annual Forecasts and Sensitivity Results

Table C.3: London-wide RQ3 forecasts for all six future waves (percent). “Exposure” is the balance in Equation 3.3, not a percentage of participants.

Wave	Persistent legacy			Legacy recovery		
	Indoor rate	Outdoor rate	Indoor exposure	Indoor rate	Outdoor rate	Indoor exposure
2023/24	12.55	9.38	57.23	12.44	8.93	58.21
2024/25	12.28	9.44	56.52	11.95	8.68	57.92
2025/26	12.27	9.43	56.54	11.85	8.61	57.94
2026/27	12.24	9.44	56.46	11.82	8.57	57.96
2027/28	12.30	9.49	56.45	11.82	8.57	57.96
2028/29	12.28	9.45	56.52	11.82	8.57	57.96

 Table C.4: ALR zero-replacement sensitivity for the frozen annual Random Forest. Test metrics are reporting-only and did not select ϵ .

ϵ	CV TV	CV MAE	Test TV	Test MAE	2028/29 exposure
10^{-4}	0.03803	0.02395	0.03635	0.02345	56.55%
10^{-5}	0.03662	0.02310	0.03723	0.02402	56.51%
10^{-6} (primary)	0.03648	0.02300	0.03700	0.02388	56.52%

The 10^{-6} primary replacement perturbs zero components least while keeping logarithms finite. The 2028/29 exposure varies by only 0.04 percentage points across the three values, so the conclusion is insensitive to this choice. Annual pooling strengths 1 and 10 change mean component estimates by approximately 0.12 percentage points relative to the primary strength 5; monthly alternatives 10 and 40 differ from the primary 20 by approximately 0.61 and 0.48 percentage points. Larger maximum differences occur only in the sparsest cells and reinforce cautious use of detailed monthly/borough results.

C.3 Monthly and Borough Diagnostics

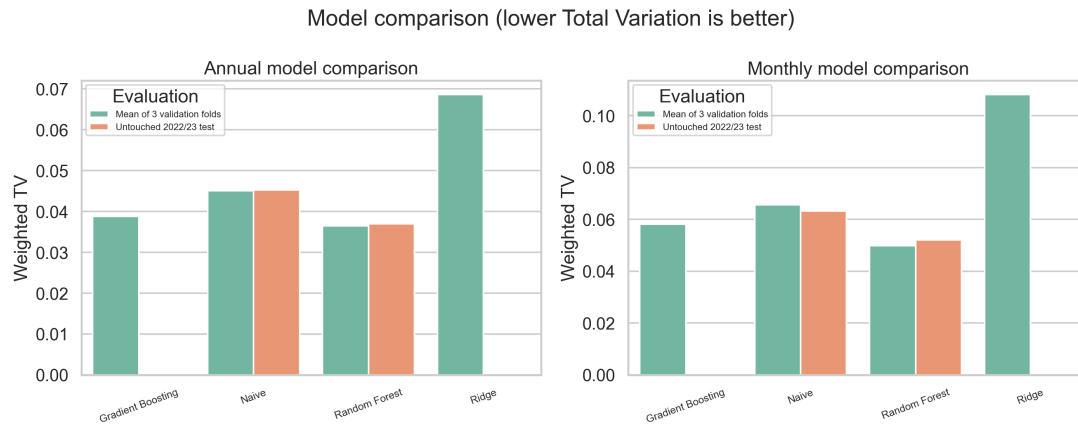


Figure C.1: RQ3 three-fold validation means and untouched 2022/23 tests. Only the validation-selected Random Forest and Naive benchmark are shown on the test split.

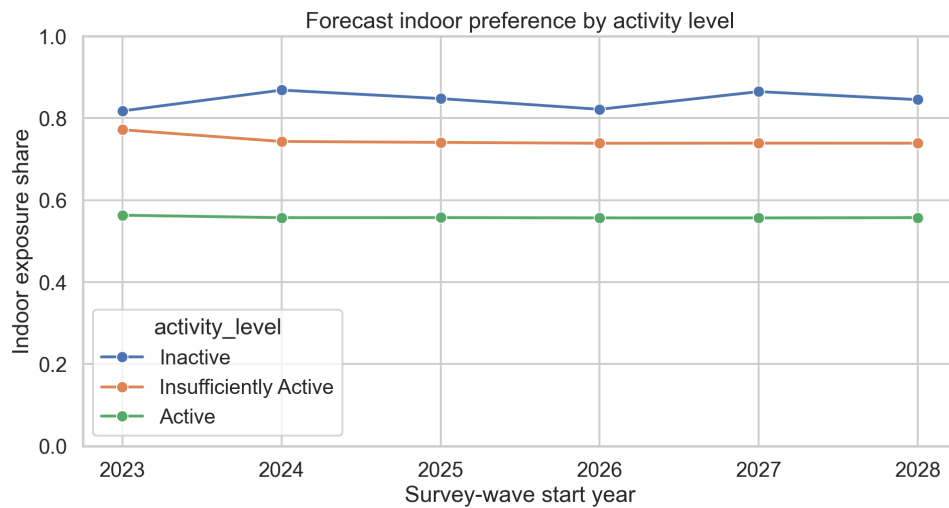


Figure C.2: Forecast indoor exposure by activity-level group under persistent legacy. Exposure is a setting balance rather than a participant proportion.

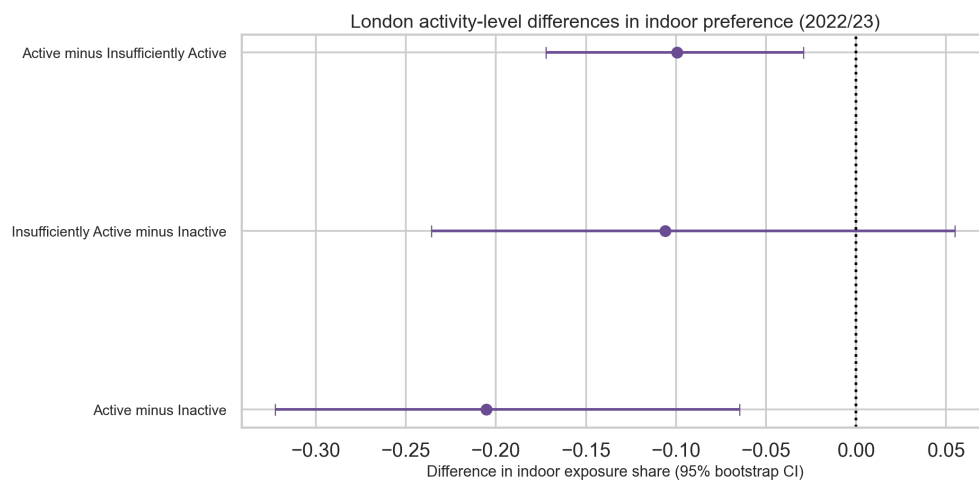


Figure C.3: Latest-wave activity-level differences in indoor exposure share. Six-wave pooled contrasts provide the primary inference.

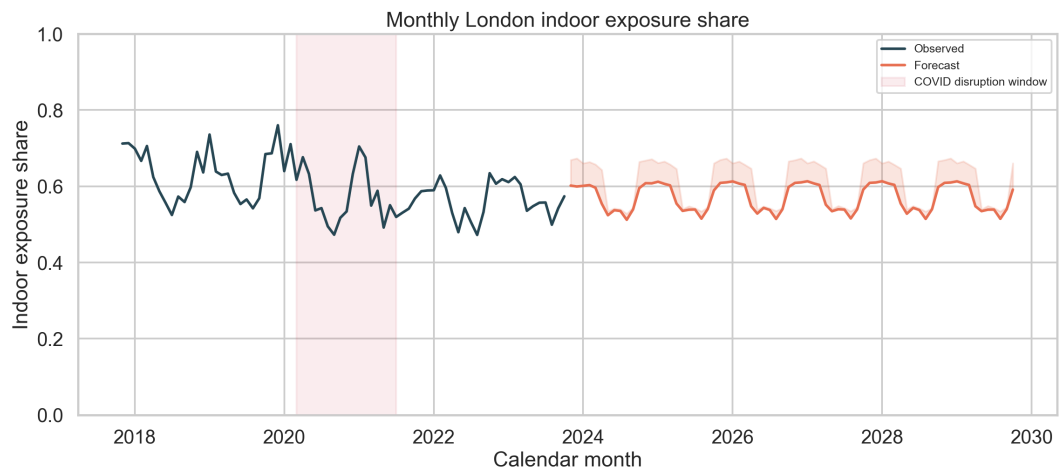


Figure C.4: Monthly London history and 2023–2028 forecasts. The validation-selected Random Forest improved on Naive in 2022/23, but extensive low-effective-sample cells make this a secondary diagnostic.

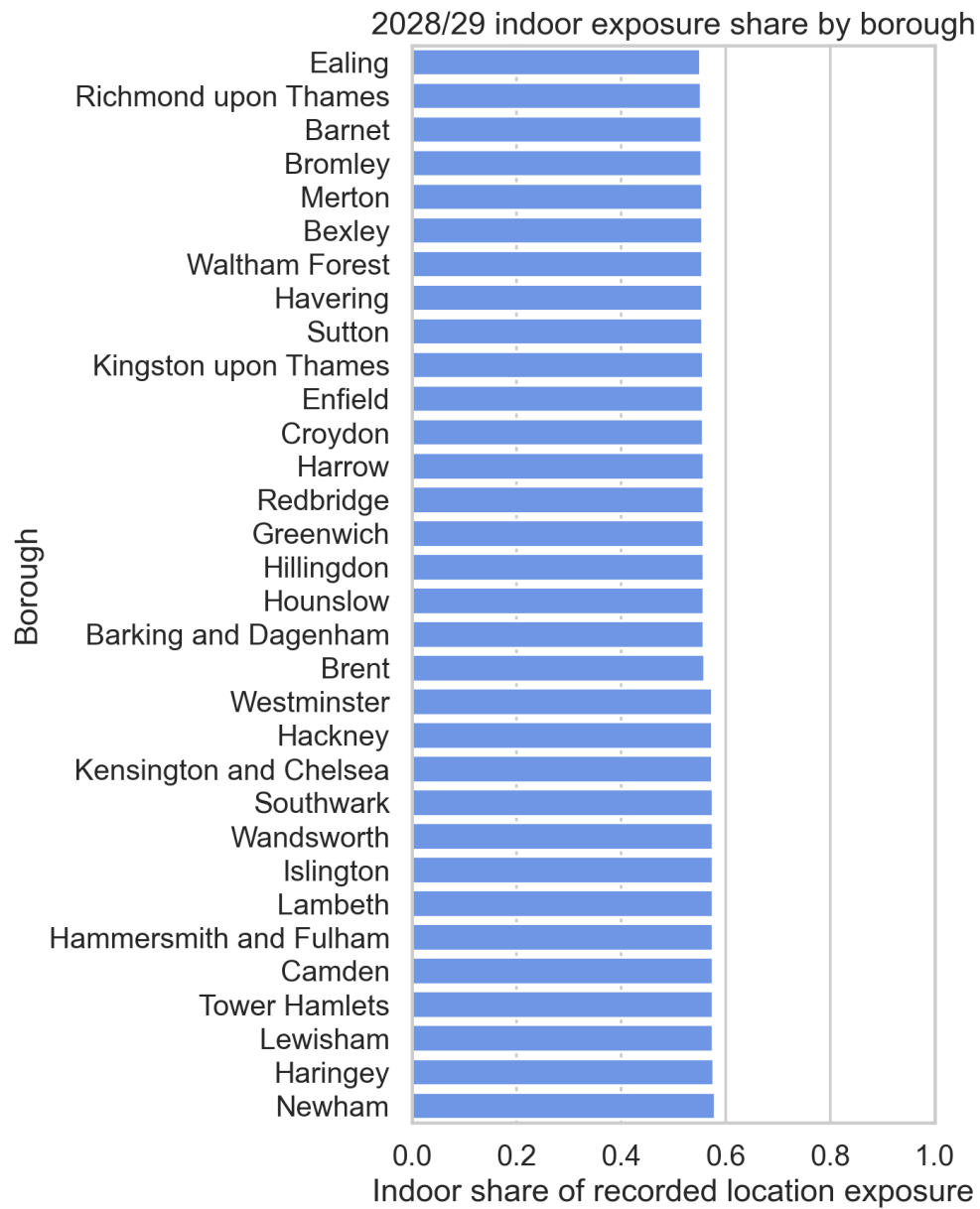


Figure C.5: Modelled borough indoor exposure in 2028/29 under persistent legacy.

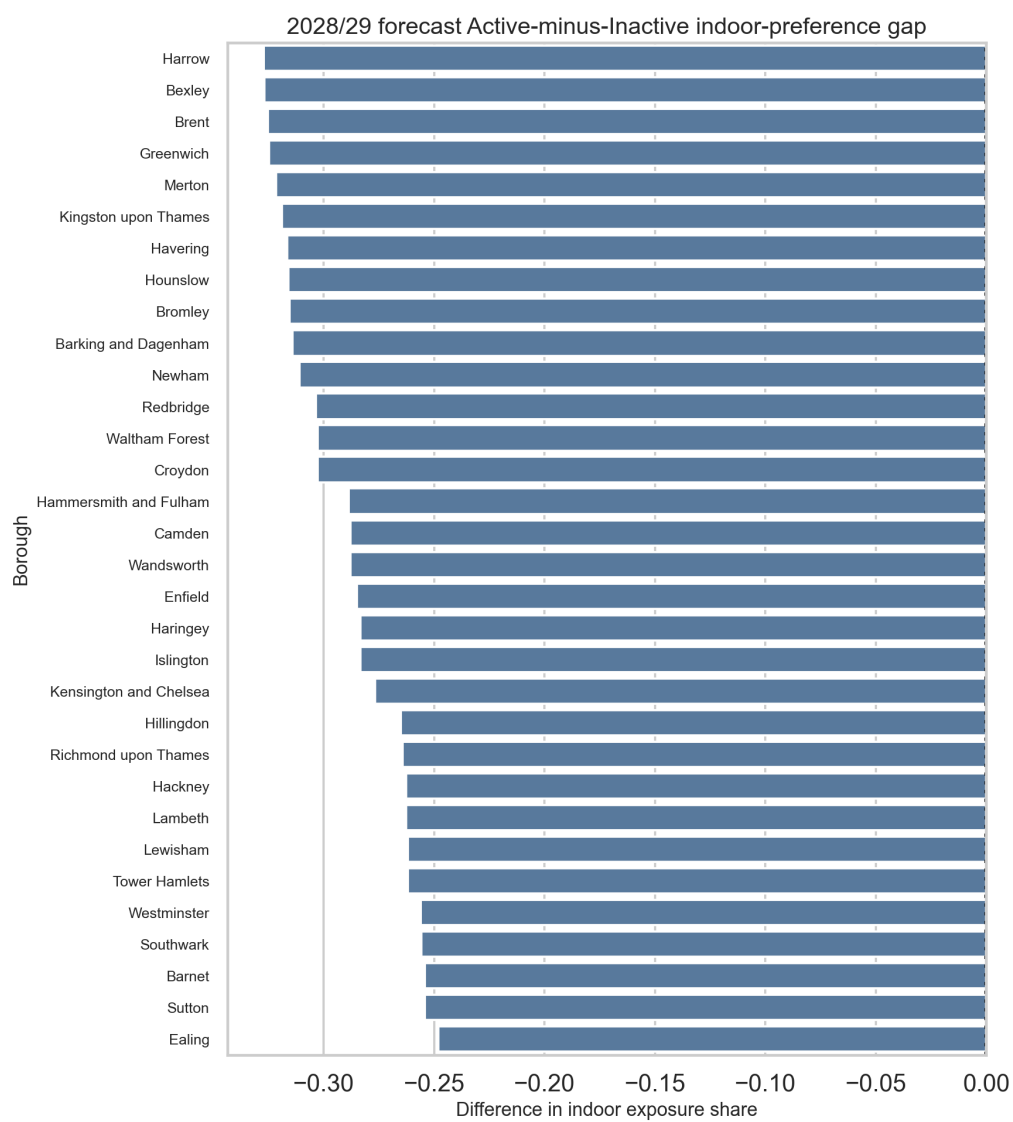


Figure C.6: Modelled borough Active–Inactive gap in 2028/29. Sparse Inactive denominators require cautious interpretation.

Appendix D

Supplementary Material for Frequent Sport Volunteering

D.1 Historical Patterns and Data Support

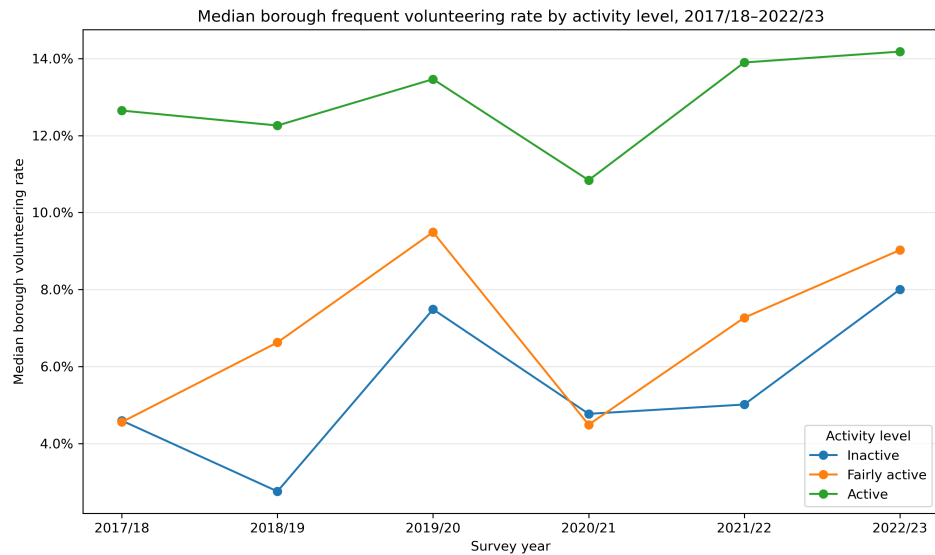
This section provides additional historical evidence underlying the RQ4 results reported in Section 4.4.

Table D.1: Historical panel coverage and sample support for RQ4.

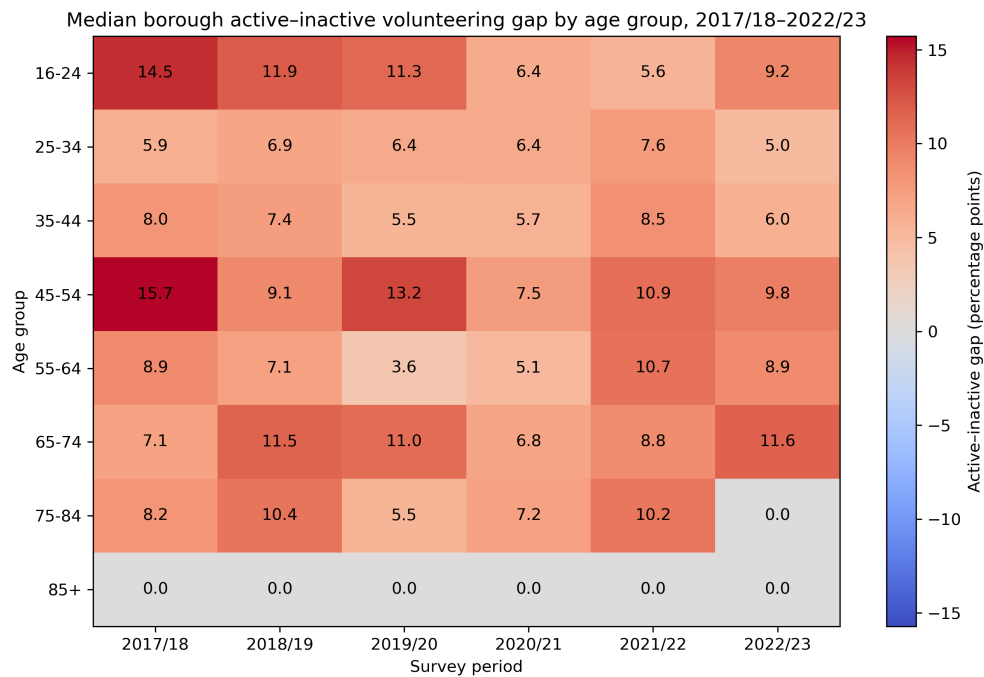
Panel	Total cells	Observed	Median n_{eff}	Small cells (%)	$n_{\text{eff}} < 5$ (%)
Borough–age–activity level	4,608	4,290	6.37	81.47	39.84
Borough–age	1,536	1,523	27.61	33.42	11.42
Borough–activity level	576	576	45.76	3.30	0.00

The decline in sample support as age and activity level are combined illustrates why the joint borough–age–activity_level results require greater caution than the two lower-dimensional panels.

Figure [D.1](#) provides the annual detail behind the historical summaries in Section 4.4.1. The activity-level ordering is visible across the observed period, while the size of the activity-level gap differs across age groups and years.



(A) Historical volunteering rates by activity level



(B) Historical age-specific activity-level gaps

Figure D.1: Supplementary historical patterns in frequent sport volunteering. Panel (A) shows annual borough-median volunteering rates by activity level from 2017/18 to 2022/23. Panel (B) shows age-specific within-borough activity-level gaps across the same historical period.

D.2 Detailed Forecast Results

This section reports the numerical values underlying the forecast patterns shown in Figures 4.8 and 4.9, together with supplementary borough-level and annual gap trajectories.

Table D.2: Projected borough-median frequent sport-volunteering rates, 2023/24–2028/29 (%).

Panel A: Age groups						
Age	2023/24	2024/25	2025/26	2026/27	2027/28	2028/29
16–24	14.69	14.69	14.69	14.69	14.69	14.69
25–34	10.21	10.21	10.21	10.21	10.21	10.21
35–44	10.21	10.21	10.21	10.21	10.21	10.21
45–54	12.22	12.22	12.22	12.22	12.22	12.22
55–64	10.21	10.21	10.21	10.21	10.21	10.21
65–74	10.21	10.21	10.21	10.21	10.21	10.21
75–84	9.49	9.49	9.49	9.49	9.49	9.49
85+	3.09	3.09	3.09	3.09	3.09	3.09

Panel B: Activity-level groups						
Activity level	2023/24	2024/25	2025/26	2026/27	2027/28	2028/29
Active	13.68	13.68	13.68	13.68	13.68	13.68
Fairly Active	8.25	7.71	8.08	7.72	8.08	7.72
Inactive	6.91	7.00	7.00	7.00	7.00	7.00

Note: Values are medians across the 32 borough projections. They are not population-weighted London-wide estimates.

Figure D.2 shows the borough-level predictions underlying the more compact borough-median summaries used in the main text.

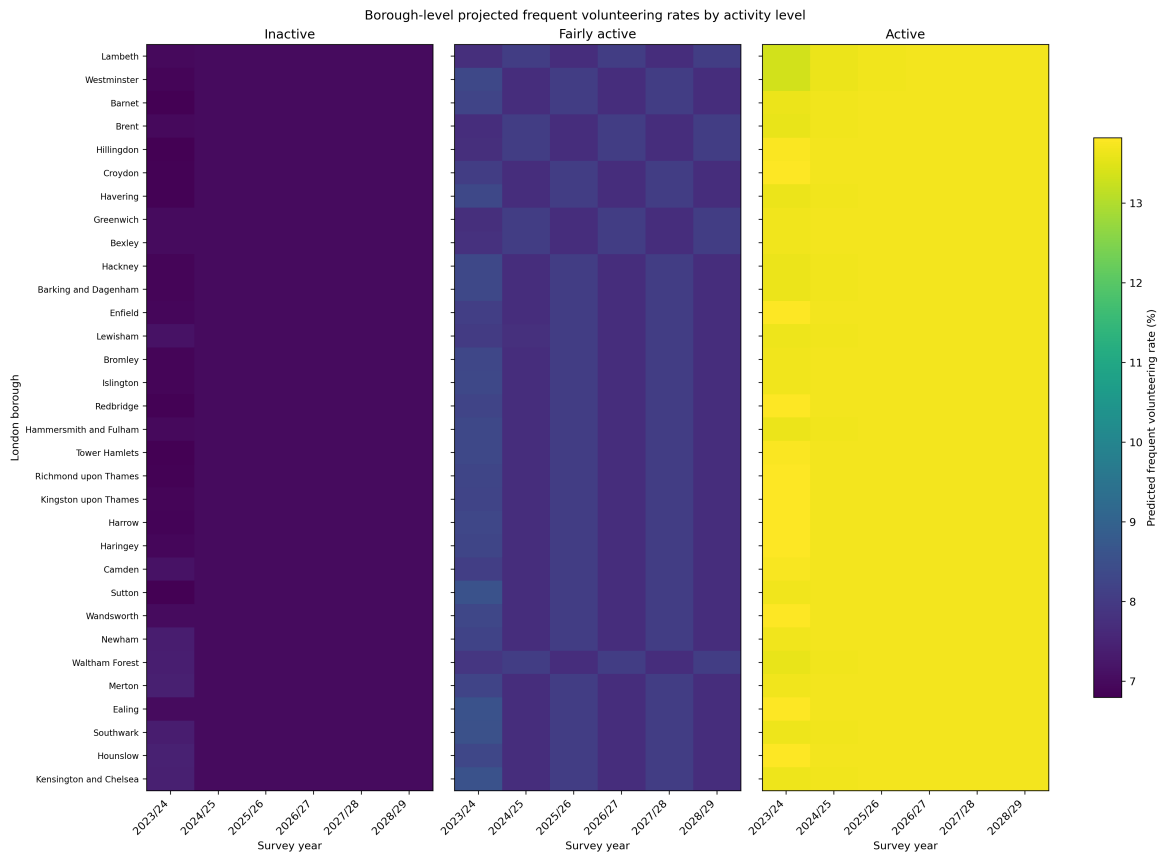


Figure D.2: Borough-level frequent sport-volunteering projections by activity level across the six forecast years.

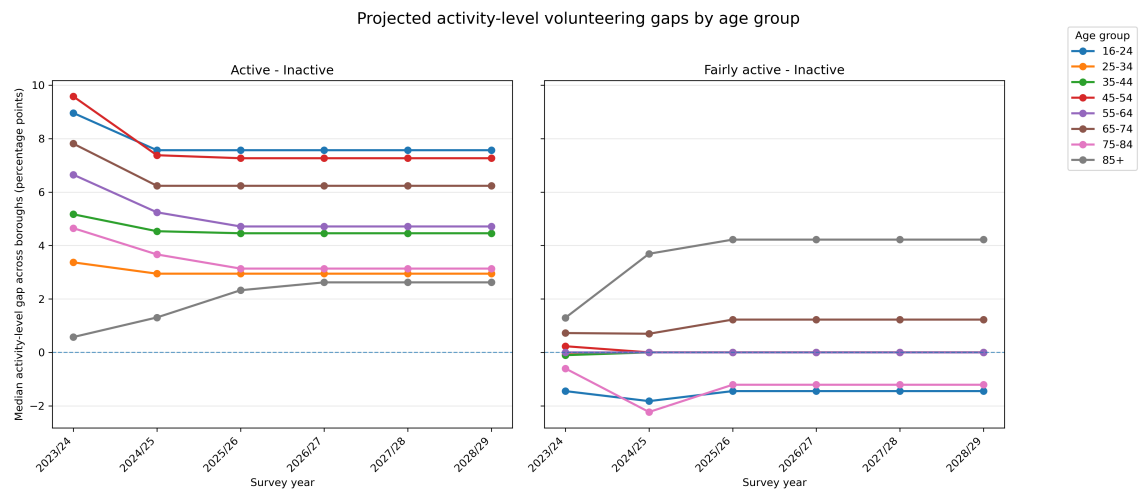


Figure D.3: Projected age-specific activity-level gaps from 2023/24 to 2028/29.

D.3 Reliability and Sensitivity

This section provides additional evidence on recursive forecast performance, variation in 2022/23 prediction error across subgroups, and the sensitivity of projected age-specific activity-level gaps to alternative model families.

Table D.3: Recursive historical backtest performance by forecast horizon.

Task	H1	H2	H3	H4
Borough–activity level	3.35	3.12	2.75	3.40
Borough–age	4.79	4.63	4.77	5.26
Borough–age–activity level	6.44	6.14	6.52	6.99
Historical forecast origins	4	3	2	1

Note: Errors are effective- n -weighted MAE in percentage points.

Table D.4: 2022/23 borough–age–activity_level forecast diagnostics by sample support and subgroup.

Panel A: Sample support		
Group	Observations	Weighted MAE (pp)
$n_{\text{eff}} < 5$	265	9.19
$5 \leq n_{\text{eff}} < 10$	170	8.89
$10 \leq n_{\text{eff}} < 30$	201	6.81
$n_{\text{eff}} \geq 30$	80	5.66
Non-small cells	148	5.84
Small cells	568	8.19

Panel B: Age groups		
Age	Observations	Weighted MAE (pp)
16–24	93	8.90
25–34	96	5.92
35–44	96	6.42
45–54	96	6.69
55–64	96	6.48
65–74	95	7.25
75–84	91	8.35
85+	53	9.94

Panel C: Activity-level groups		
Activity level	Observations	Weighted MAE (pp)
Active	246	6.27
Fairly Active	225	10.08
Inactive	245	7.06

The sensitivity results distinguish the two gap measures. The Active–Inactive gap is relatively robust in direction for most 16–84 age groups, particularly by the final forecast year, although Ridge Regression produces larger differences in magnitude than Random Forest. The Fairly Active–Inactive contrast shows more frequent sign reversals. However, these reversals often occur where the primary gap is close to zero, so a change in sign does not necessarily indicate a large numerical difference. The 85+ group remains more model-dependent for both contrasts, consistent with its weaker historical sample support.

Table D.5: Sensitivity of age-specific activity-level gaps to alternative model families.

Panel A: Median absolute difference from the primary Gradient Boosting gap (pp)										
Alternative	Year	Gap	16–24	25–34	35–44	45–54	55–64	65–74	75–84	85+
Random Forest	2023/24	Active–Inactive	1.54	1.82	1.51	2.33	1.60	1.24	0.42	0.86
Random Forest	2023/24	Fairly Active–Inactive	2.01	1.05	0.82	1.35	0.88	0.92	0.68	2.17
Random Forest	2028/29	Active–Inactive	0.64	0.74	0.80	0.62	0.54	0.95	1.51	2.58
Random Forest	2028/29	Fairly Active–Inactive	1.69	0.08	0.28	0.22	0.20	0.60	1.19	4.37
Ridge Regression	2023/24	Active–Inactive	8.80	3.44	4.98	9.10	6.16	7.32	4.50	0.55
Ridge Regression	2023/24	Fairly Active–Inactive	1.87	0.47	0.91	1.26	0.27	0.95	0.62	1.60
Ridge Regression	2028/29	Active–Inactive	7.42	2.84	4.34	7.07	4.58	6.06	2.99	2.60
Ridge Regression	2028/29	Fairly Active–Inactive	1.47	0.03	0.03	0.05	0.03	1.16	1.23	4.22

Panel B: Borough-level gap sign-reversal rate (%)										
Alternative	Year	Gap	16–24	25–34	35–44	45–54	55–64	65–74	75–84	85+
Random Forest	2023/24	Active–Inactive	0.0	0.0	0.0	0.0	0.0	0.0	0.0	28.1
Random Forest	2023/24	Fairly Active–Inactive	62.5	15.6	25.0	31.2	18.8	31.2	56.2	50.0
Random Forest	2028/29	Active–Inactive	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.4
Random Forest	2028/29	Fairly Active–Inactive	100.0	0.0	0.0	3.1	0.0	0.0	46.9	100.0
Ridge Regression	2023/24	Active–Inactive	12.5	31.2	6.2	9.4	3.1	9.4	12.5	25.0
Ridge Regression	2023/24	Fairly Active–Inactive	46.9	6.2	21.9	9.4	6.2	3.1	59.4	84.4
Ridge Regression	2028/29	Active–Inactive	0.0	0.0	0.0	0.0	0.0	0.0	0.0	12.5
Ridge Regression	2028/29	Fairly Active–Inactive	100.0	0.0	0.0	3.1	0.0	0.0	100.0	100.0

Note: The primary borough–age–activity_level specification is Gradient Boosting. Panel A reports the median absolute borough-level difference between an alternative-model gap and the corresponding primary gap. Panel B reports the percentage of the 32 boroughs in which the alternative model changes the sign of the primary gap.